

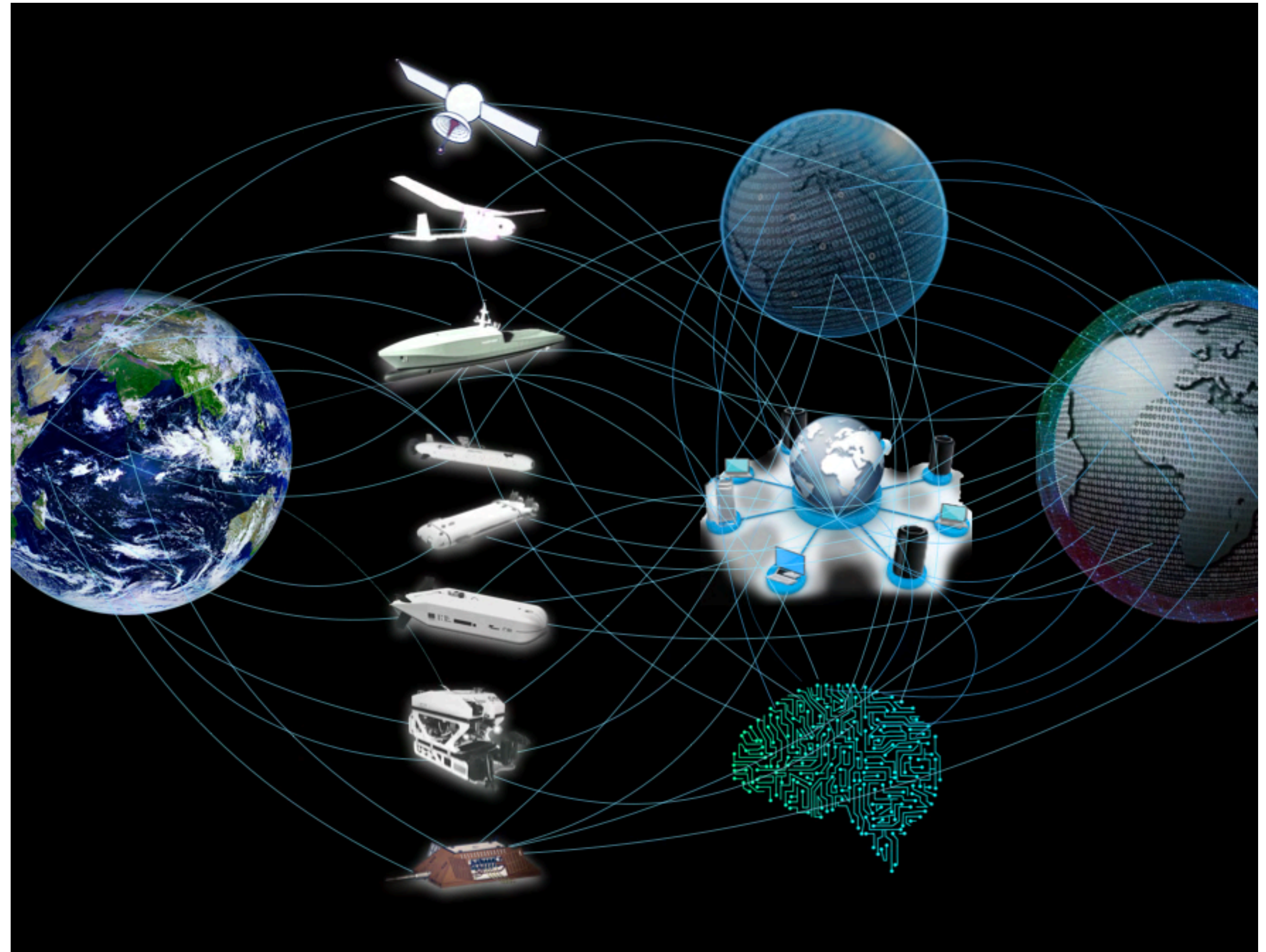
쉽게 쓸 수 있는 해양데이터 플랫폼

해양 속으로 스며드는 해양 정보



차 례

1. 일상 속에 해양 정보
2. 디지털 트윈 & 메타버스
3. 해양정보 플랫폼
4. 국내 해양 메타버스 현황



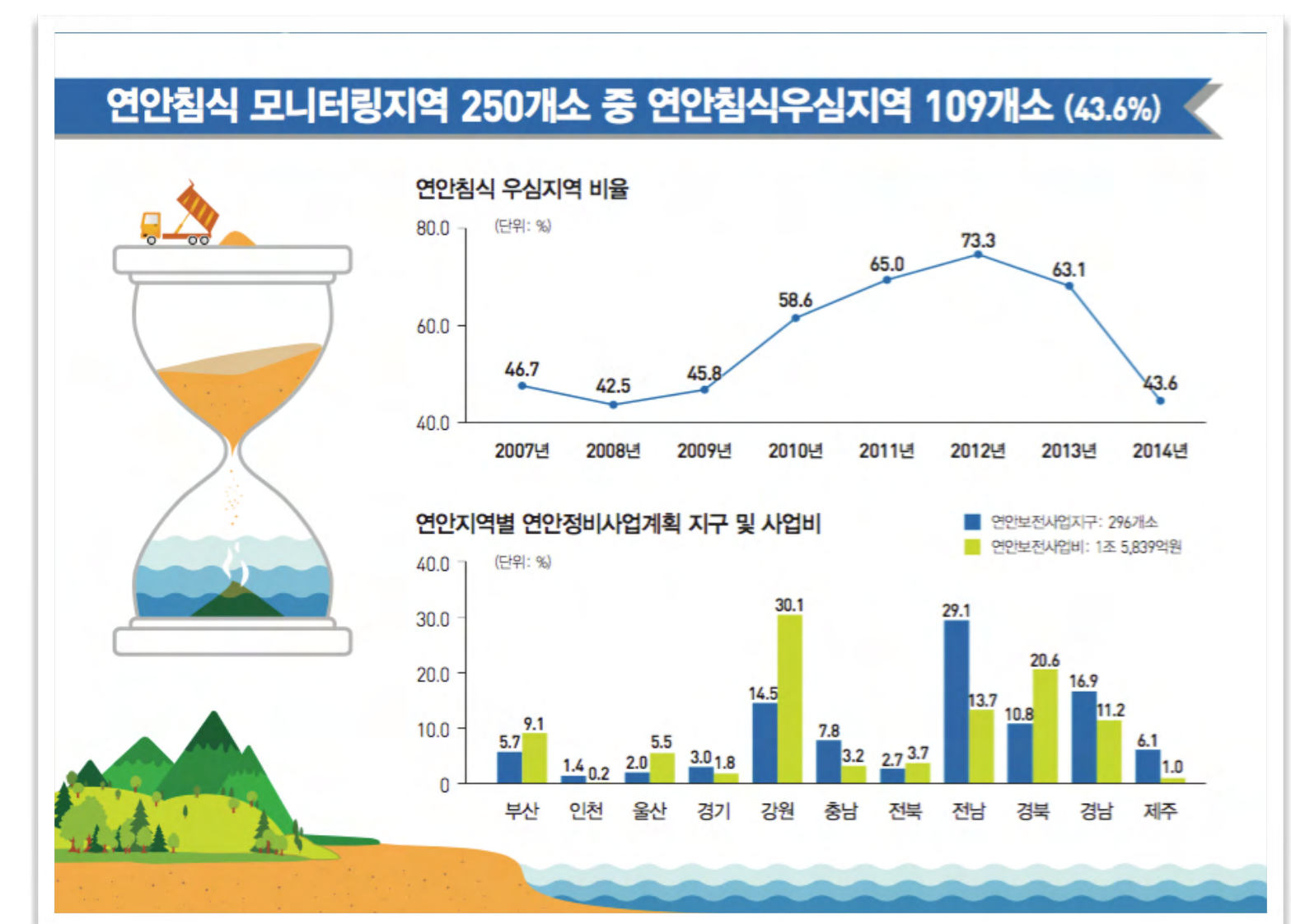
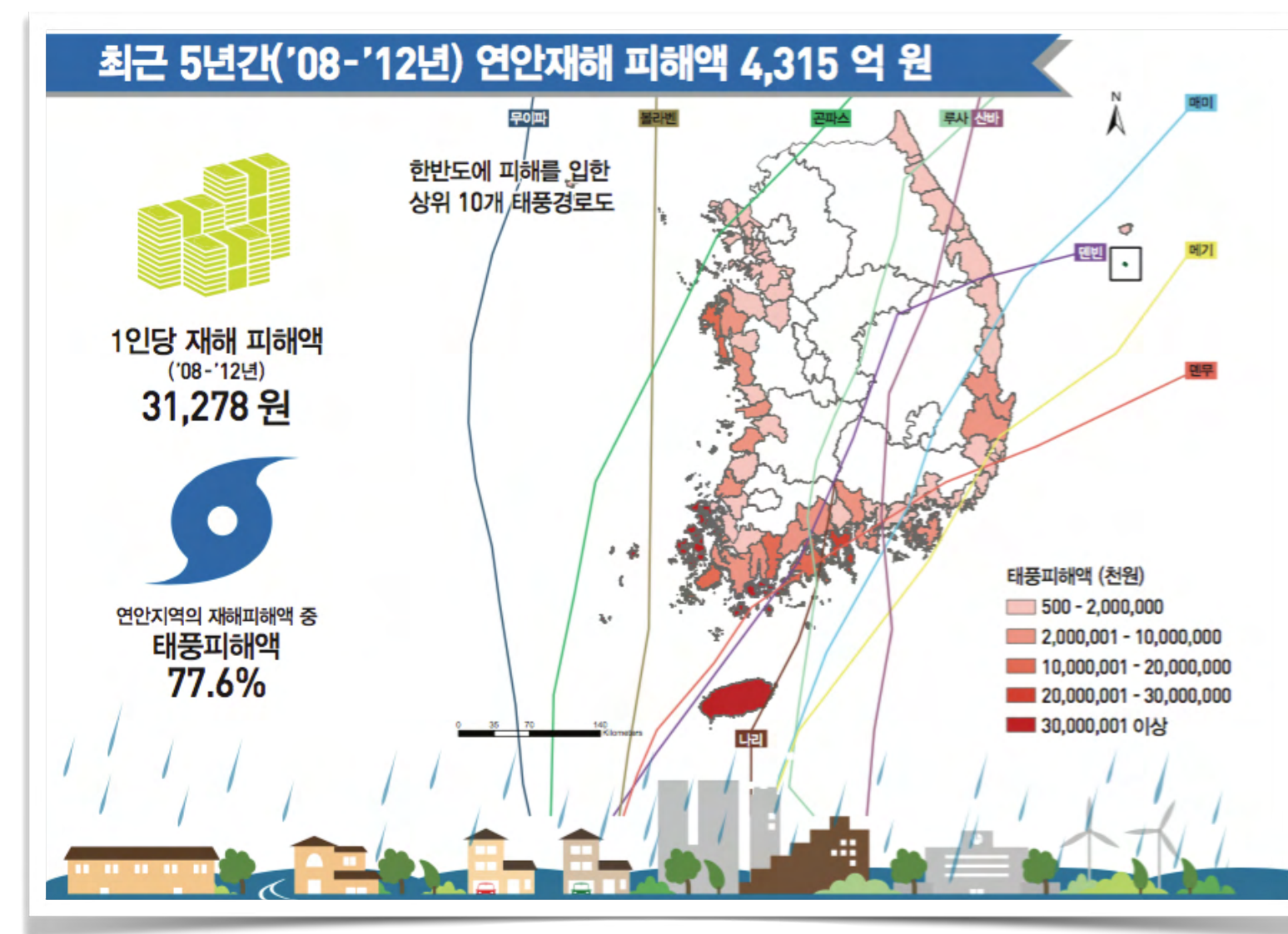
1. 일상 속에 해양 정보

일상 속에서의 해양 정보

우리나라 해양영토 = 4.4 X 육지영토

연안재해

연안침식



(국립해양조사원 자료)

생활 속에서의 해양 데이터

인류 생존 이슈

중장기

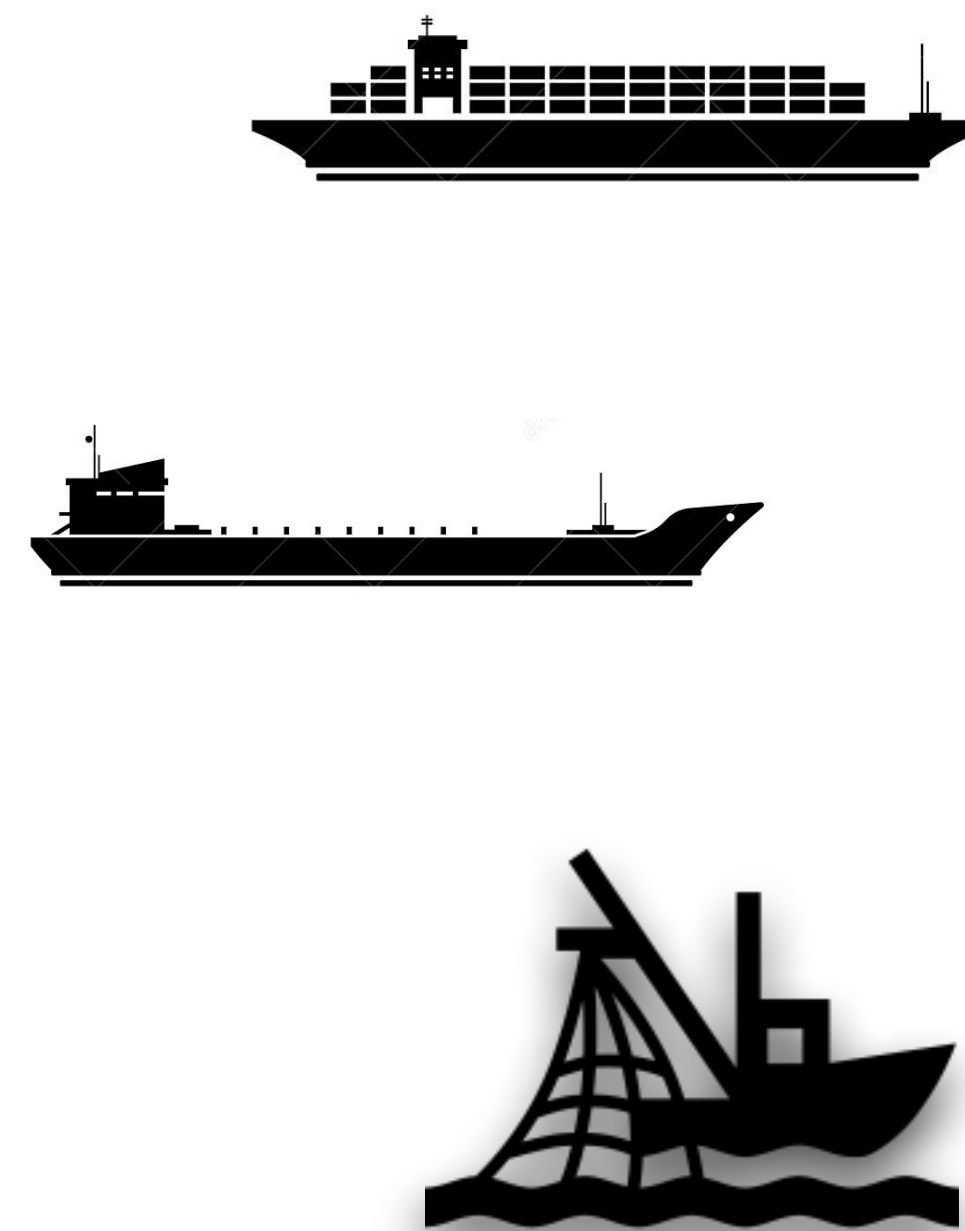
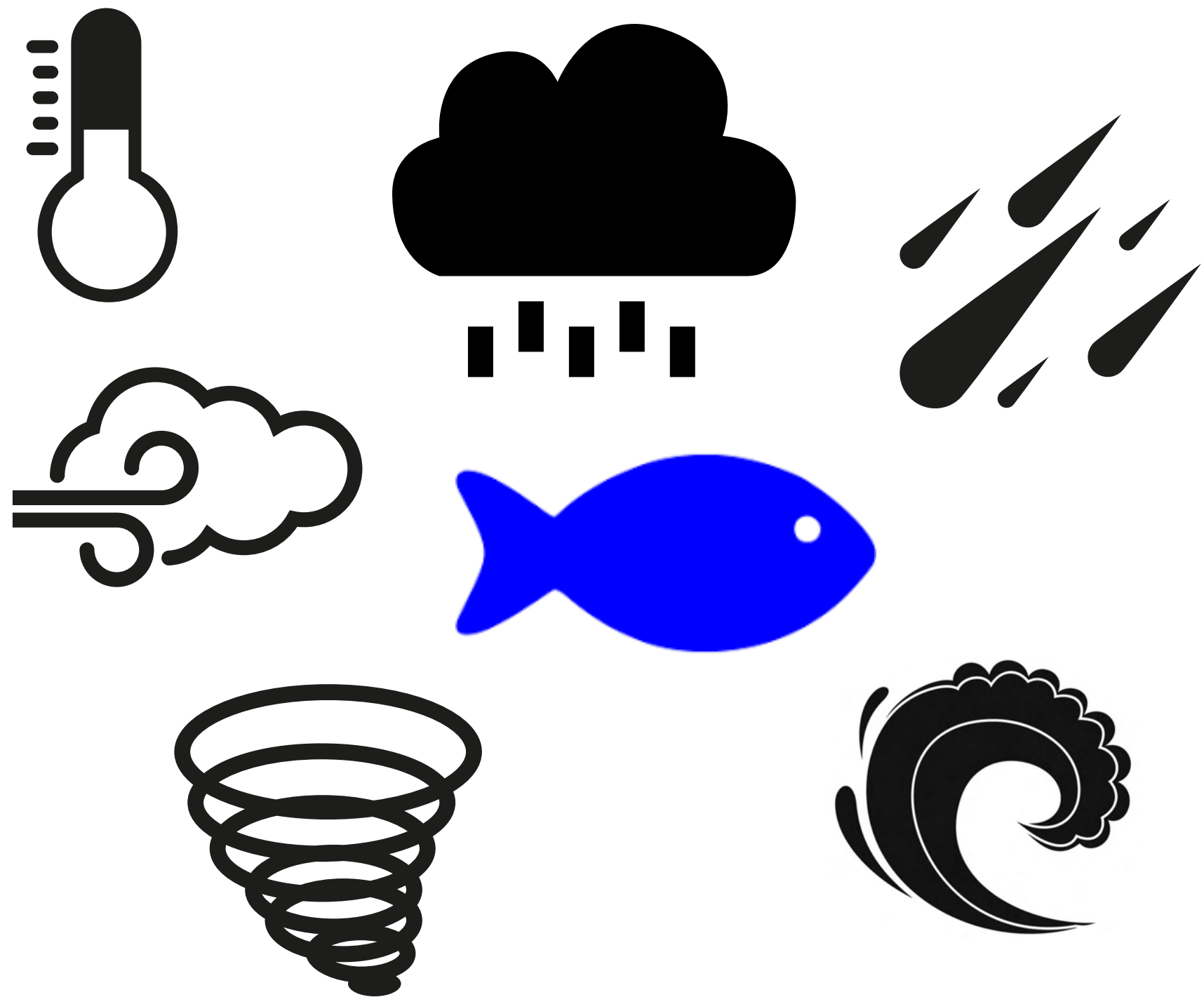
일상

기후 변화

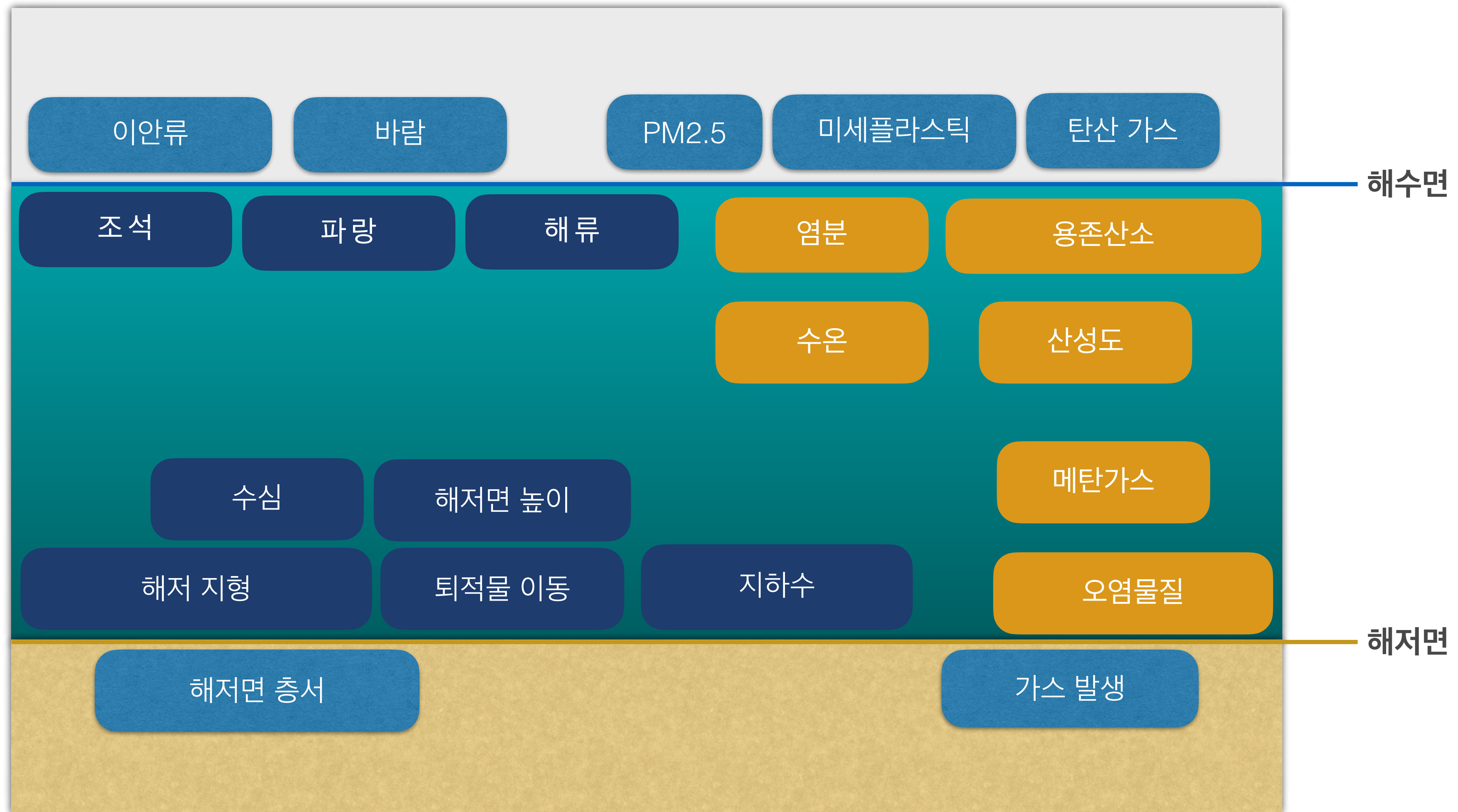
미래 (해양)
식량자원

산업·경제 활동

여가 활동

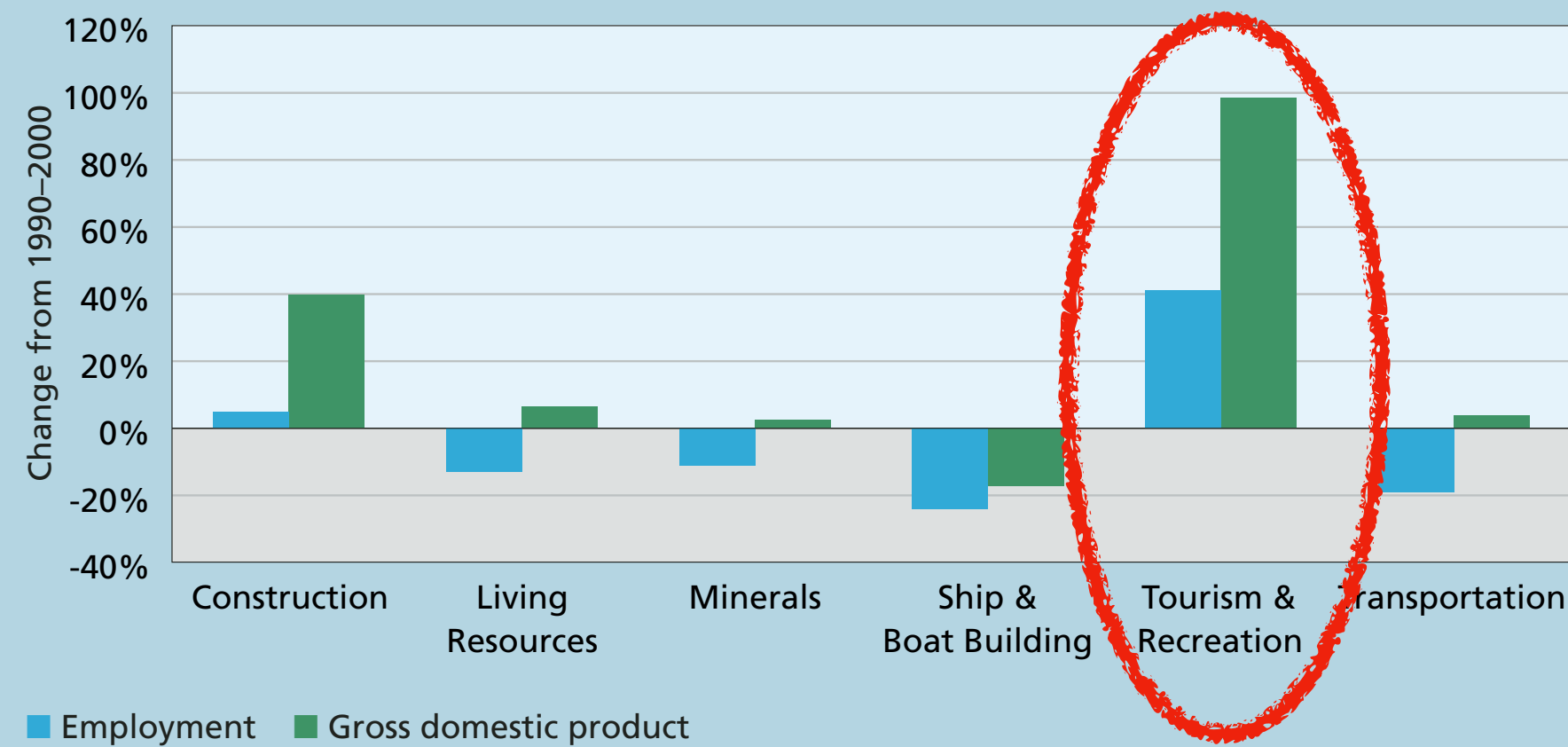


● 해양정보: 기후변화 및 바다생태환경 정보



An Ocean Blueprint for the 21st Century: final report (US Commission on Ocean Policy, 2004)

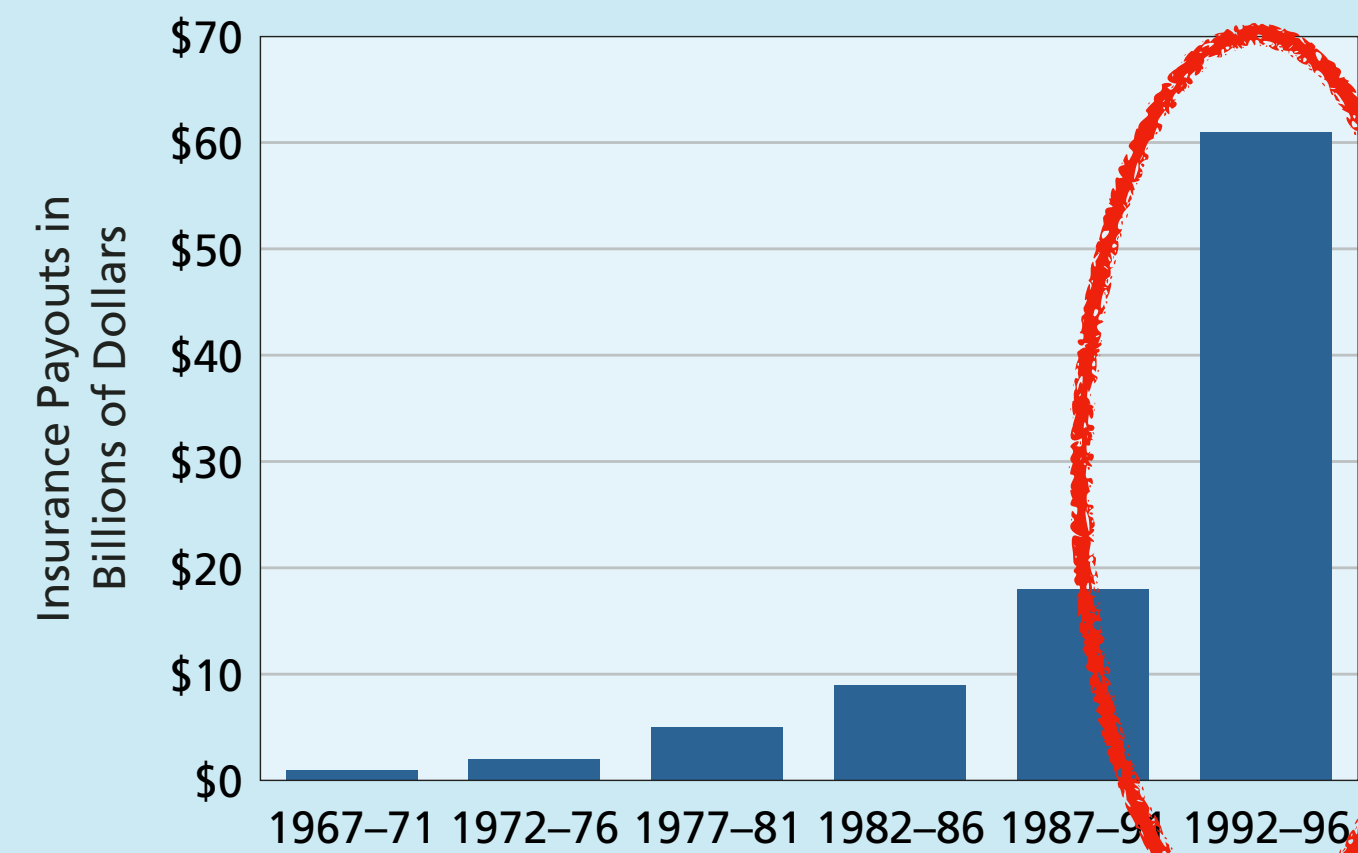
Figure 1.4 The Shift from Goods to Services in the Ocean Economy



Between 1990 and 2000, the ocean economy experienced a significant increase in the importance of service-oriented activities. This trend is clearly illustrated by the dramatic increase in both employment and output associated with tourism and recreation. Shifts in employment and revenue in the traditional goods-producing sectors—minerals, living resources, transportation, ship and boat building—were affected by changes in technology, national priorities, and the status of living and nonliving resources.

Source: Living Near... and Making a Living From... the Nation's Coasts and Oceans, Appendix C.

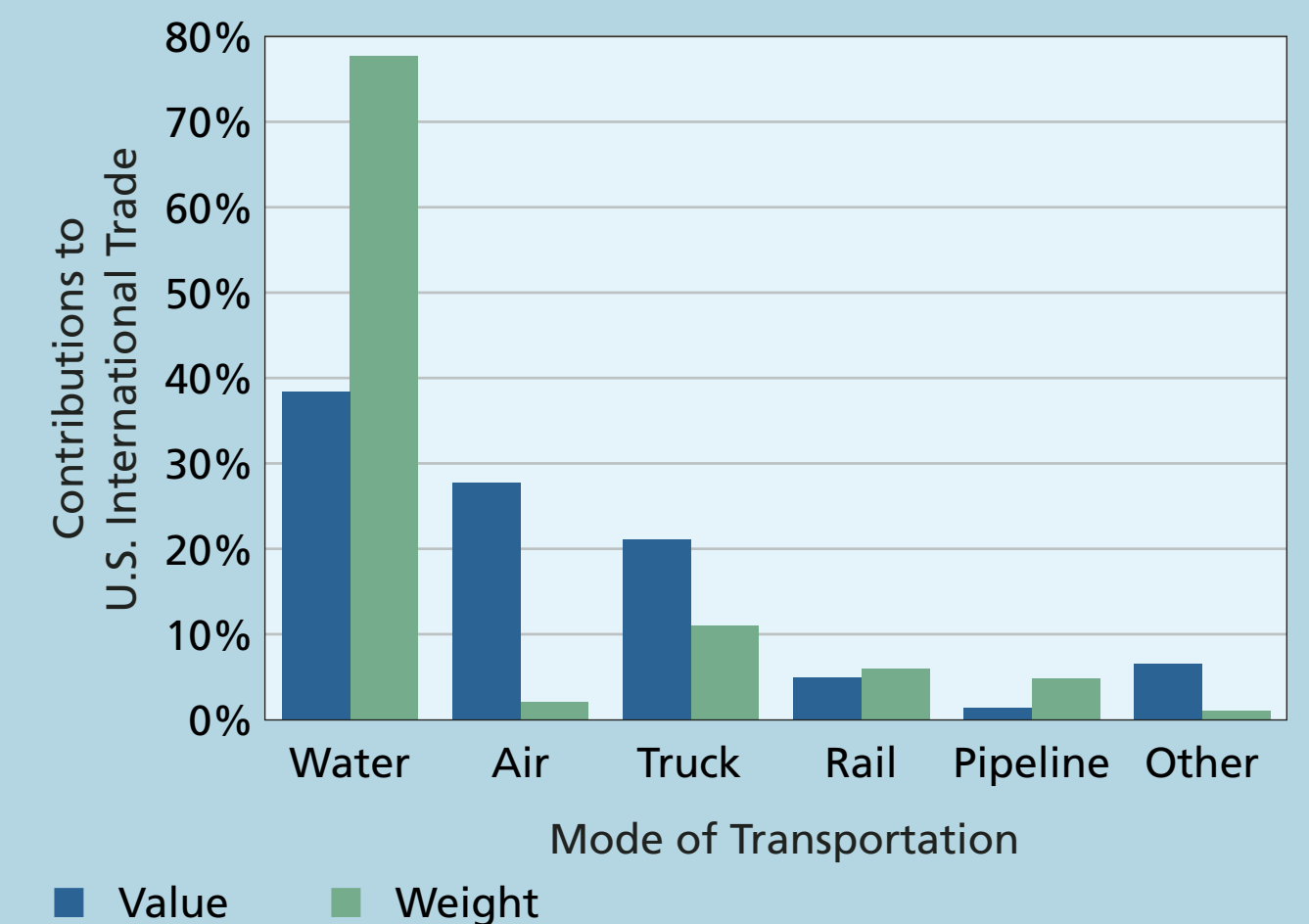
Figure 10.1 The Growing Cost of Natural Disasters



In the thirty years between 1967 and 1996, insurance companies have experienced a 6,000 percent increase in payouts to federal and private insurance holders for damages due to natural catastrophes.

Source: Consumer Federation of America. *America's Disastrous Disaster System*. Washington, DC, January 1998.

Figure ES.8 Ports are the Primary Gateway for International Trade



In 2001, U.S. ports were major gateways for international trade. Waterborne commerce accounted for 78 percent of total U.S. international trade by weight (1,643 million tons) and 38 percent by value (\$718 billion).

Source: U.S. Department of Transportation, Bureau of Transportation. "U.S. International Trade and Freight Transportation Trends 2003." <www.bts.gov/publications/us_international_trade_and_freight_transportation_trends/2003/> (Accessed May 2004).

An Ocean Blueprint for the 21st Century: final report (US Commission on Ocean Policy, 2004)

Flood of Ocean & Coastal Data

Figure 28.1 The Flood of Ocean and Coastal Data into NOAA



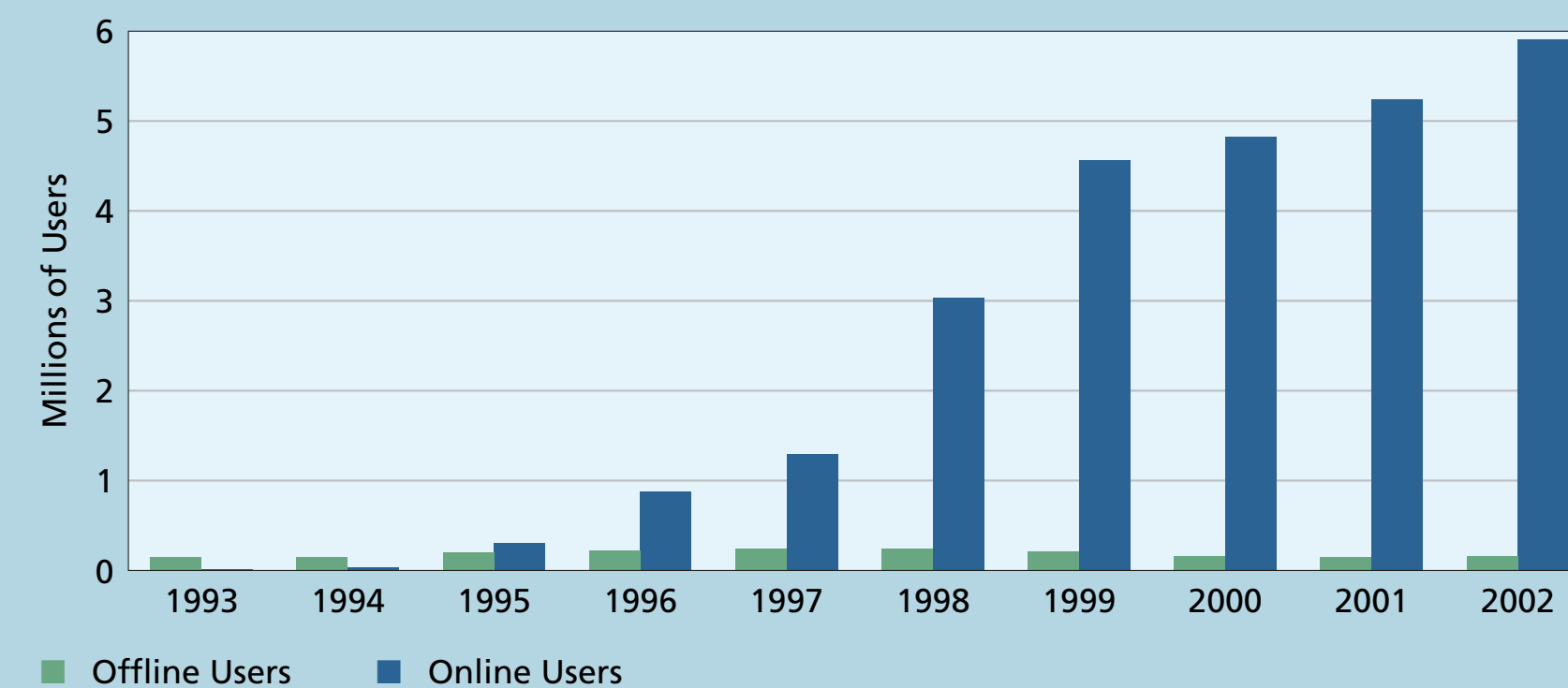
Between 2002 and 2017, NOAA's data holdings are expected to grow by a factor of 100, to 74 million gigabytes. (One gigabyte equals about one billion bytes; one terabyte equals about one thousand gigabytes.)

Source: National Oceanic and Atmospheric Administration. *The Nation's Environmental Data: Treasures at Risk: A Report to Congress on the Status and Challenges for NOAA's Environmental Data Systems*. Washington, DC: U.S. Department of Commerce, 2001.

현재 보다 수 백배의 해양 데이터 생산 예상

2017년에는 74,000 TB 의 해양 데이터 확보 예측

Figure 28.2 The Growing Demand for Ocean Data

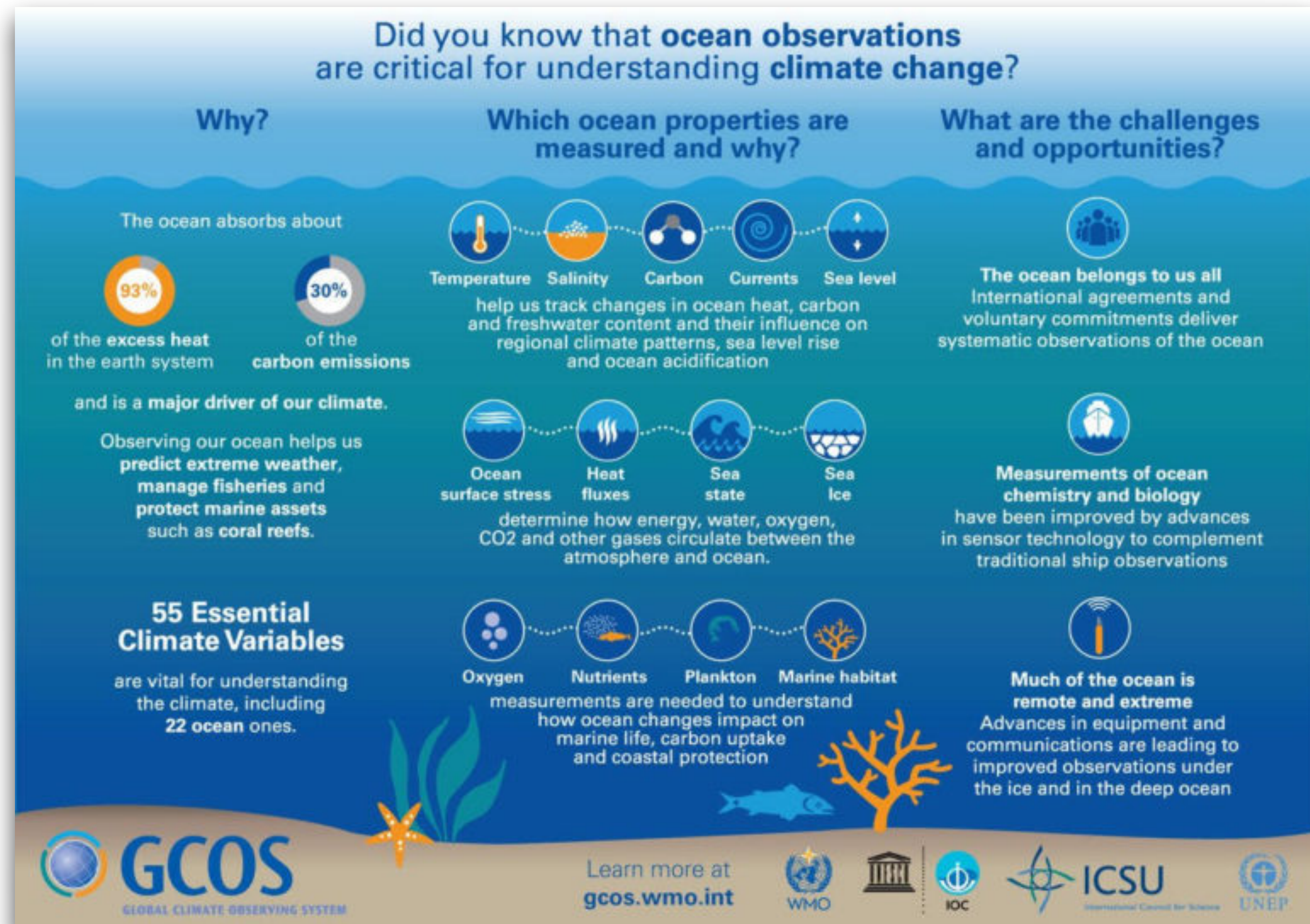


Online users are requesting increasing amounts of environmental data and information from NOAA each year. Improved data management practices are needed to address the growing volume of requests.

Source: National Oceanic and Atmospheric Administration. *The Nation's Environmental Data: Treasures at Risk: A Report to Congress on the Status and Challenges for NOAA's Environmental Data Systems*. Washington, DC: U.S. Department of Commerce, 2001.

오프라인보다 온라인에서의 해양자료 수요가 폭발적으로 증가하고 있음

● 해양정보: 기후변화 및 바다생태환경 정보



Ocean & Coastal Science Needs I: Issues

기후변화 예측 및 대응

해양 생태계 기능
(해양식량자원)

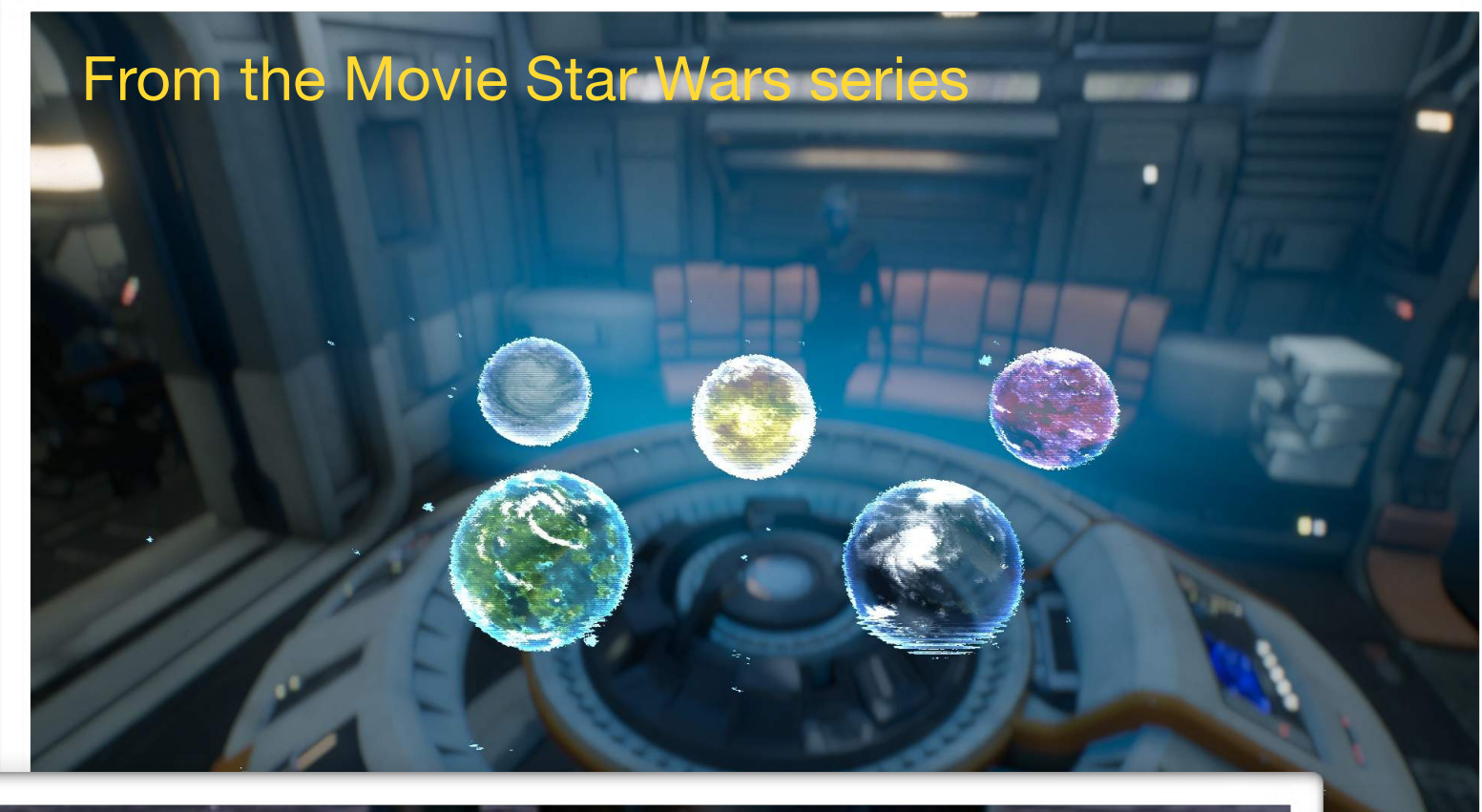


빅데이터 ➡ 이야기를 갖고 있는 **살아있는 데이터**

빅데이터
문제점 ➡ 너무 많아서 보기 어렵다

그 많은 데이터를 어떻게 **빠르게 & 쉽게**
보고 & 활용할 것인가?

디지털 트윈 / 메타버스

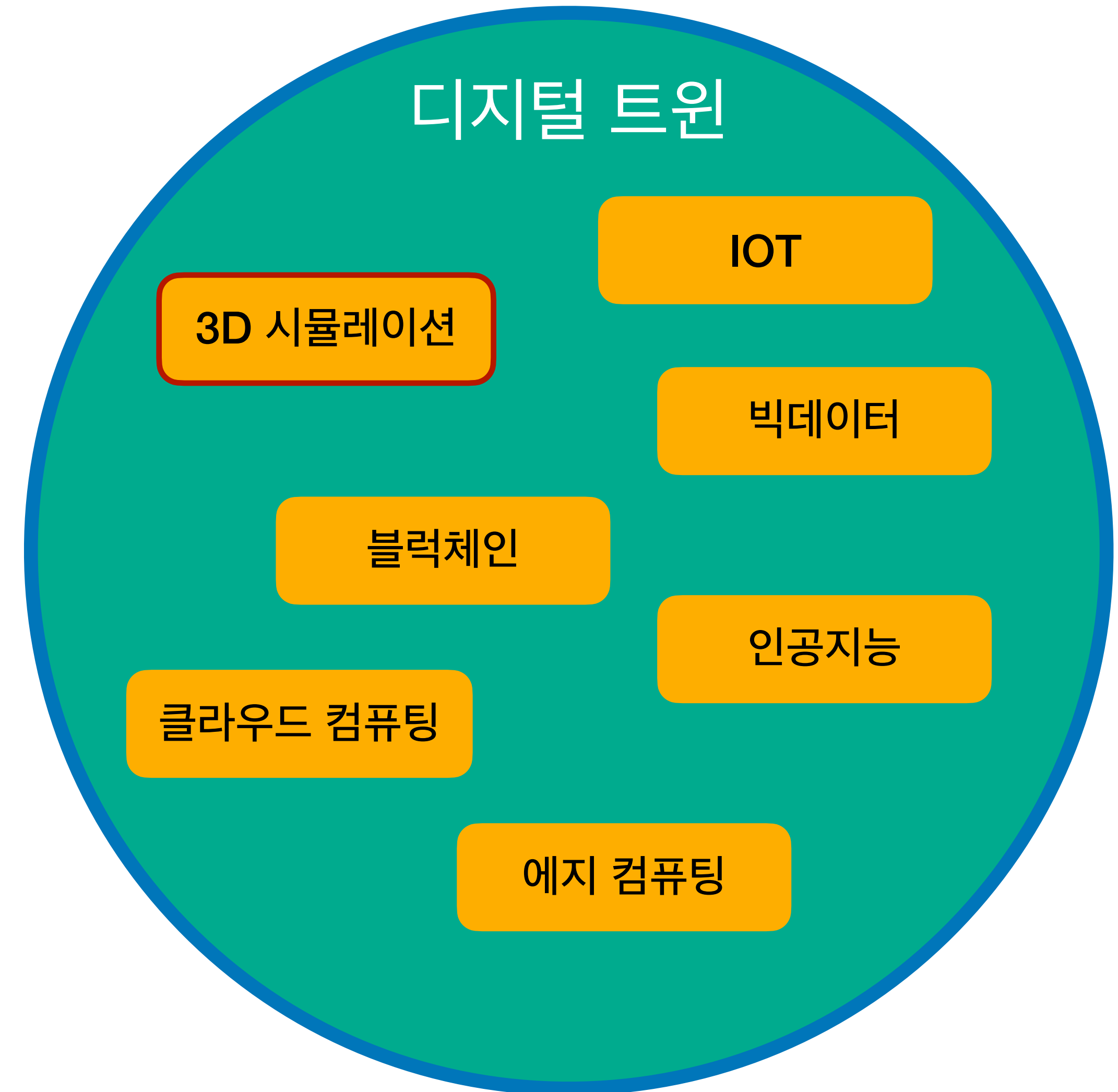


2. 디지털 트윈 & 메타버스

제4차산업혁명? 디지털 트윈? 메타버스?



개념 (Concept)



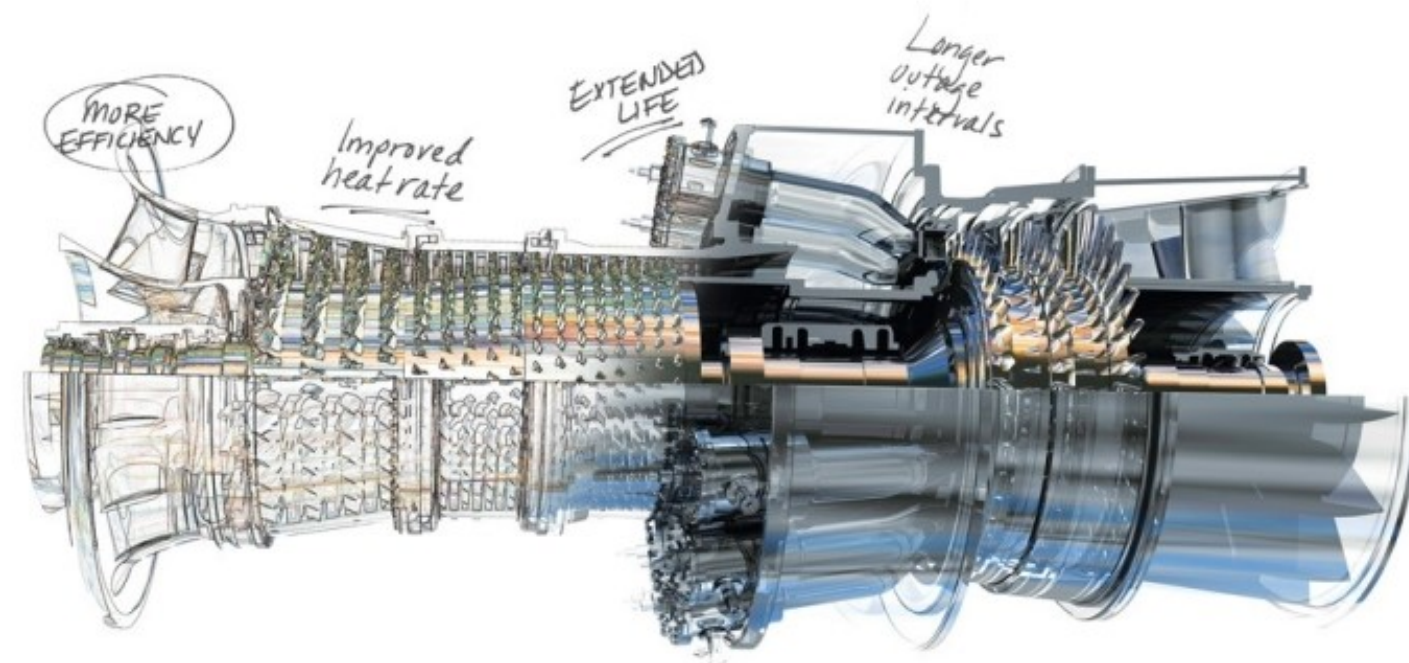
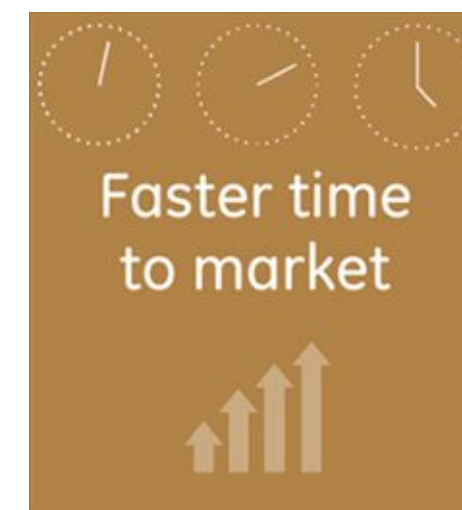
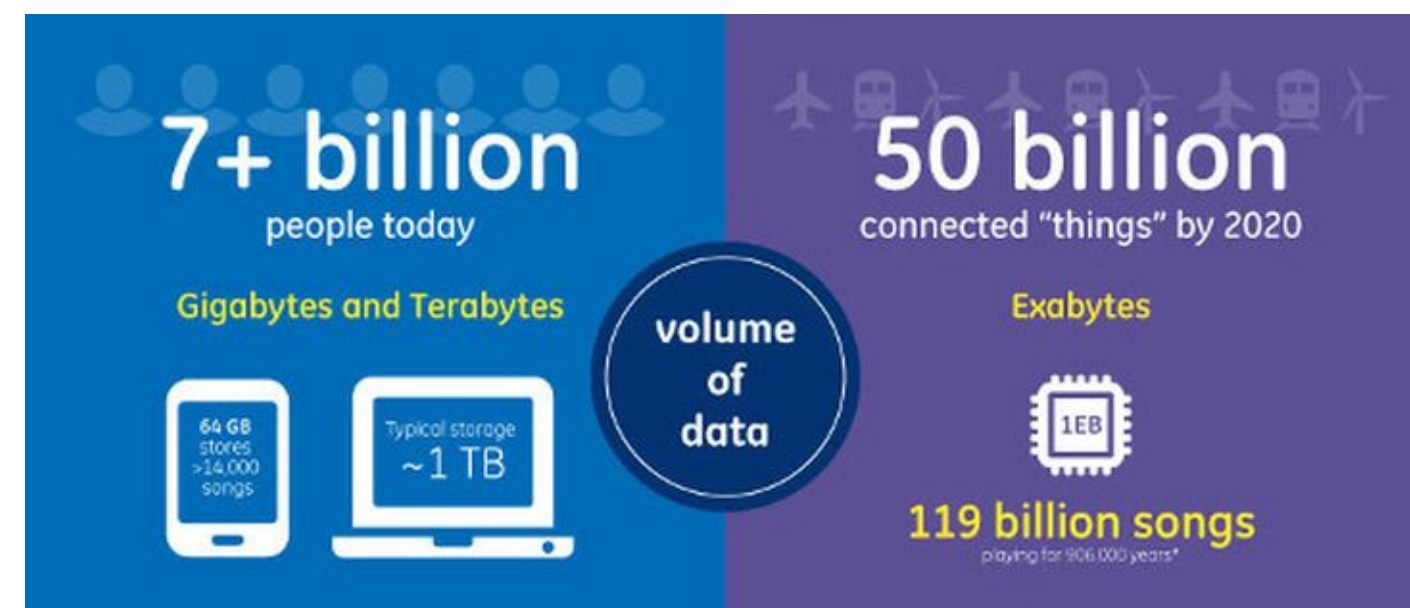
DIGITAL TWIN (Smart Factory)



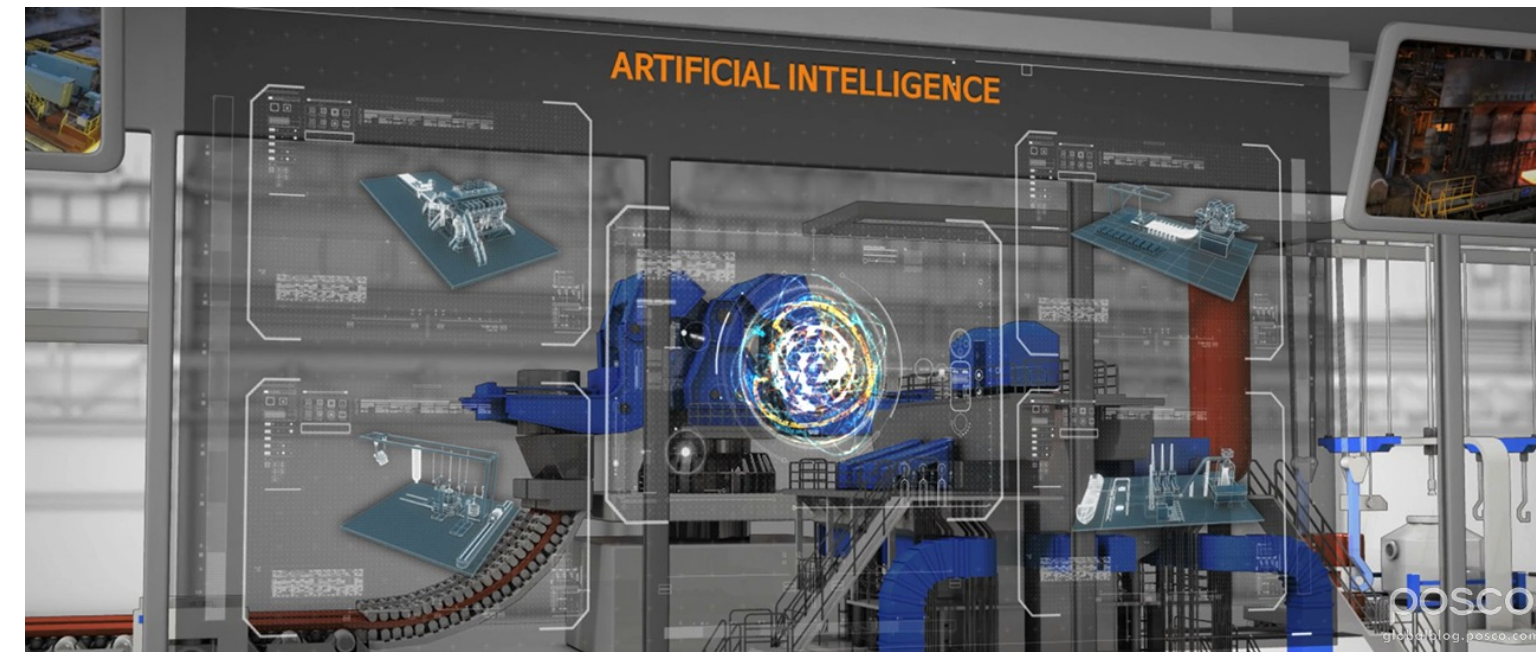
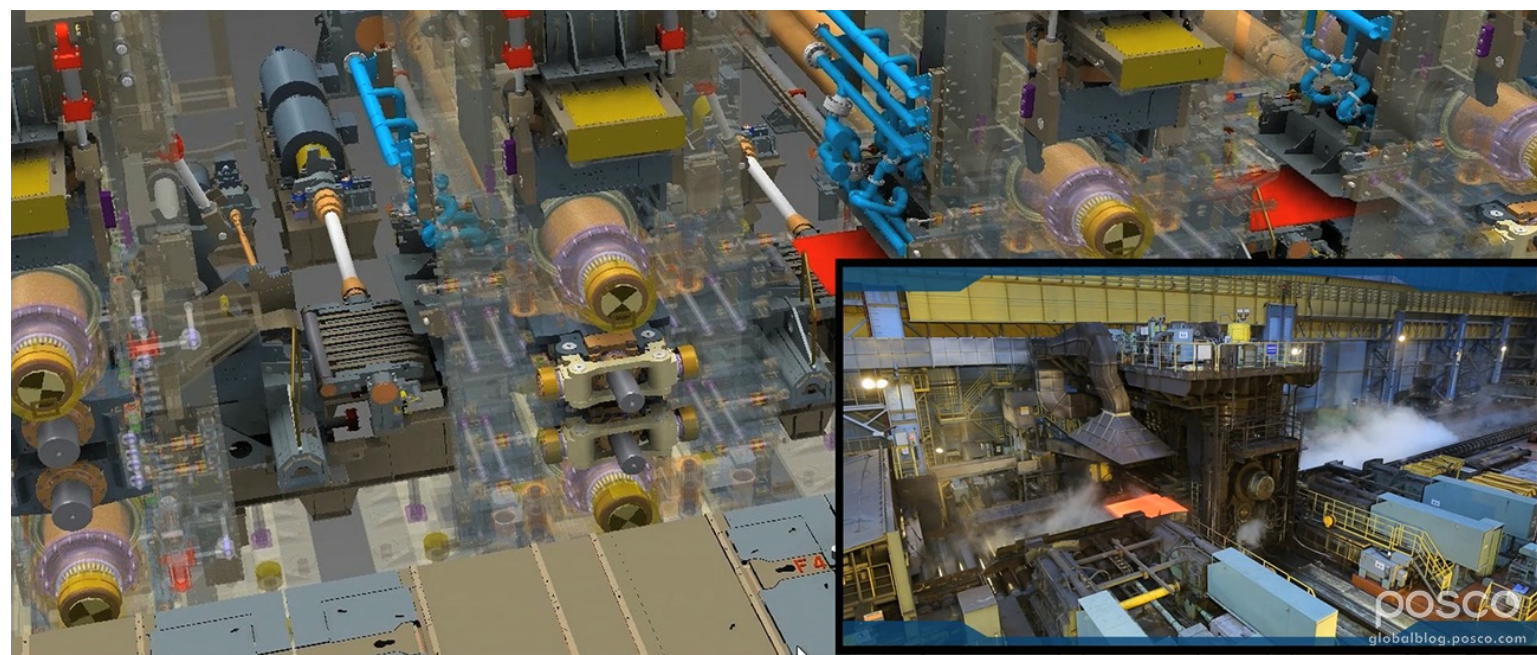
GE Imagination at work

<https://www.ge.com/digital/>

GE PREDIX Cloud Platform (Industrial Internet Platform)



POSCO's Smart Factory

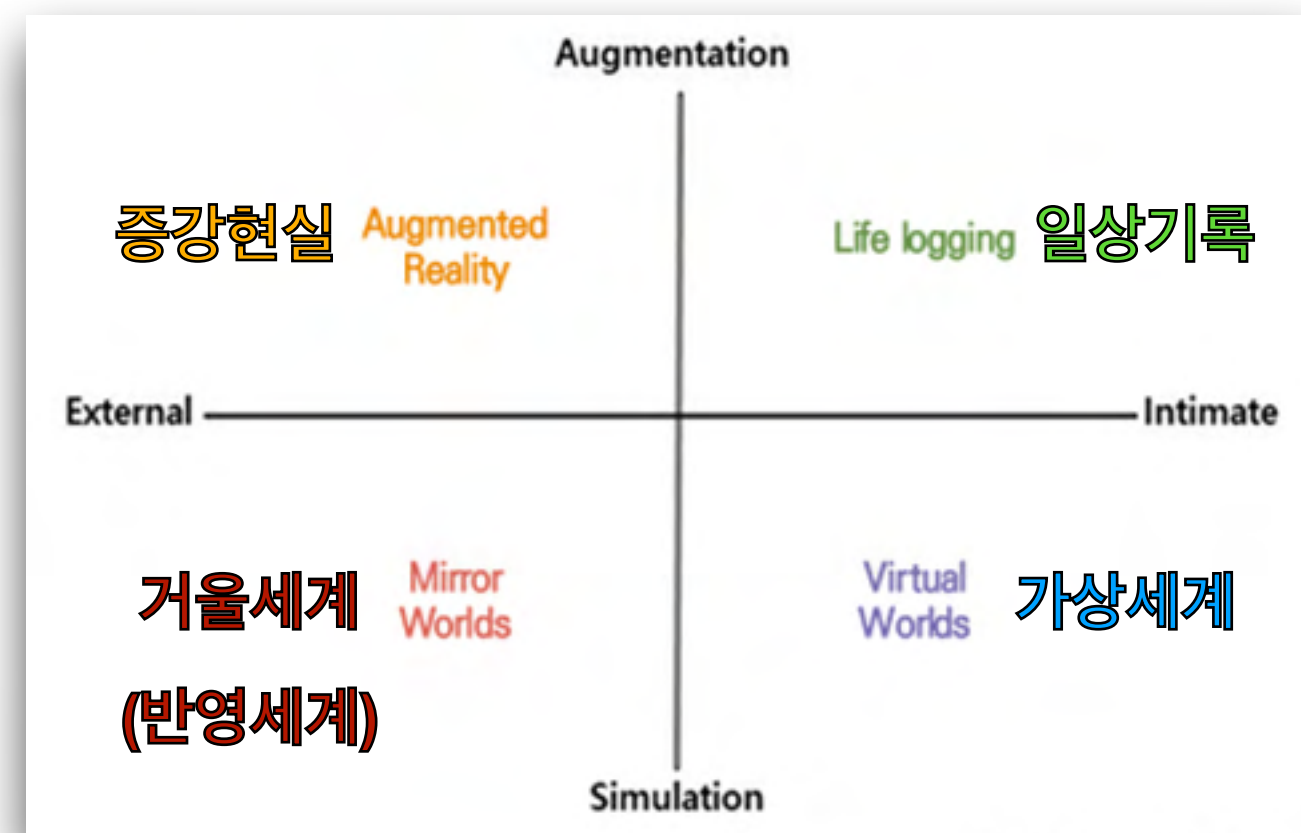


메타버스의 부상

COVID-19 덕분?

메타버스 (Metaverse)

가상을 의미하는 '메타(meta)'와 세계를 의미하는 '유니버스(universe)'의 합성어로 가상과 현실 세계가 상호작용하는 3차원 가상세계



메타버스의 분류 (ASF*, 2007)

*Acceleration Studies Foundation

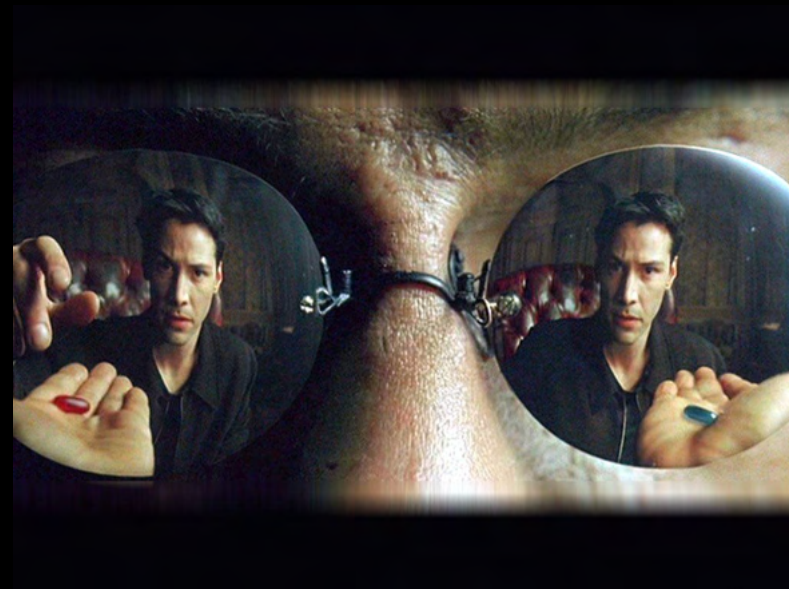
online

VR·AR·XR

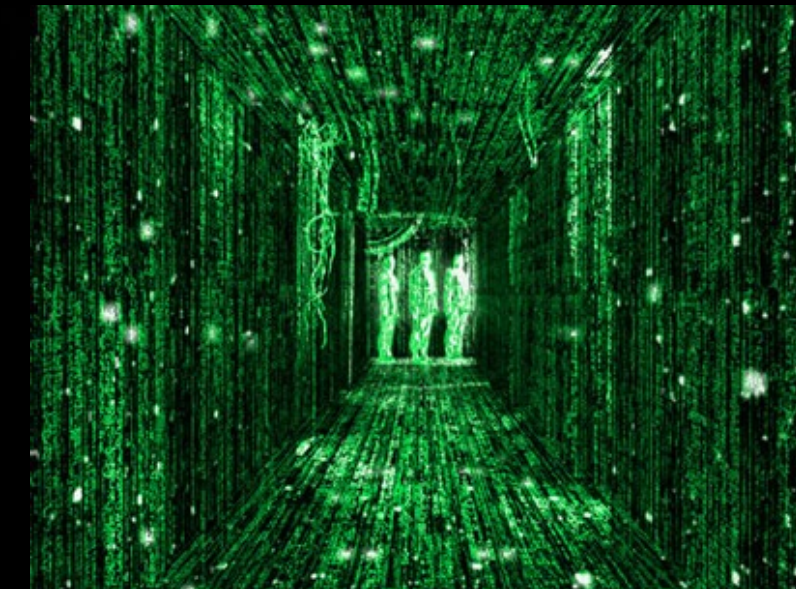


The Matrix (1999)

Real World



Dream World



가상현실 대규모집단 온라인 롤 플레이 게임
(VRMMORPG: Virtual reality massively multiplayer online role-playing game)

소드 아트 온라인
(SAO: Sword Art Online, 2002)



이세계(異世界) : 우리가 살고 있는 세계와는 다른 세계를 말하며 일본의 만화, 애니메이션과 같은 미디어 장르로 주로 사용
(위키백과, 2021)

별천지?



‘OASIS’ 가상현실 시뮬레이션(virtual reality simulation)

레디 플레이어 원
(Ready Player One, 2018*)



* 원작 소설: Ernest Cline, 2011

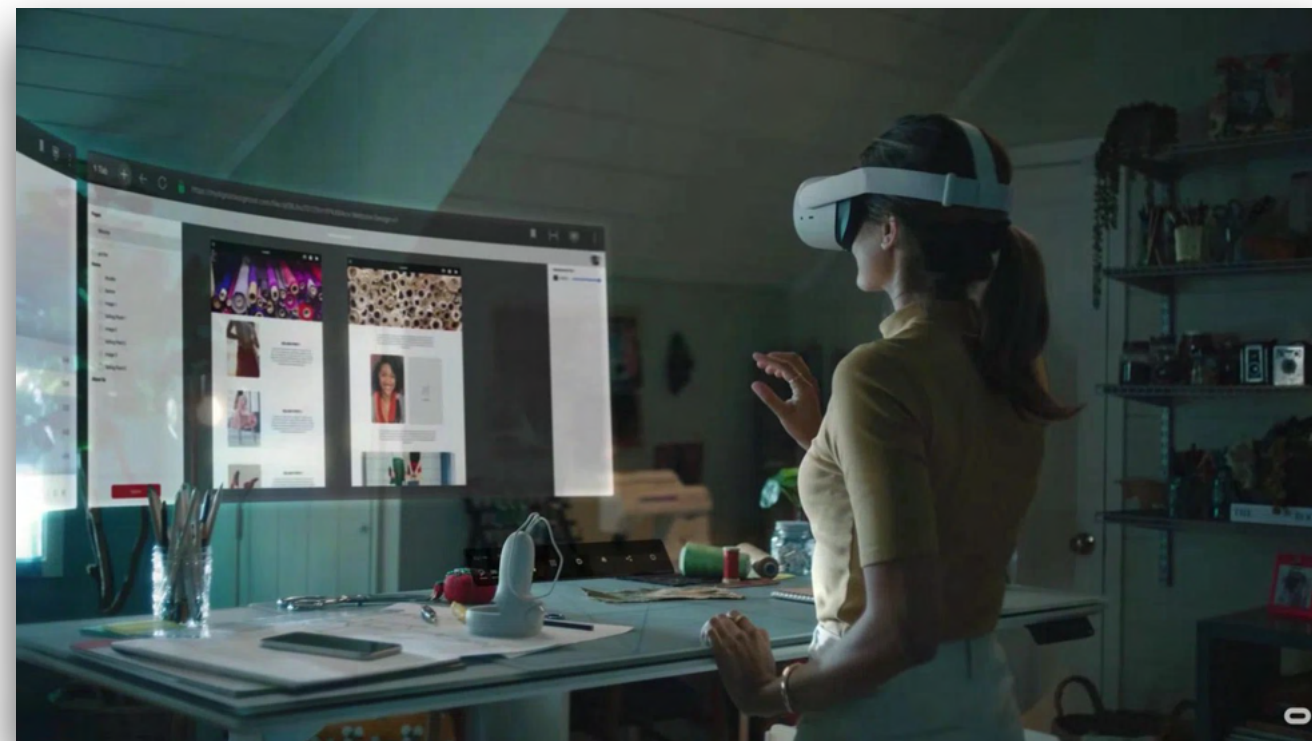
메타(Meta)*

*구 페이스북(facebook)

가상현실(Virtual Reality)
소셜 플랫폼(Social Platform)



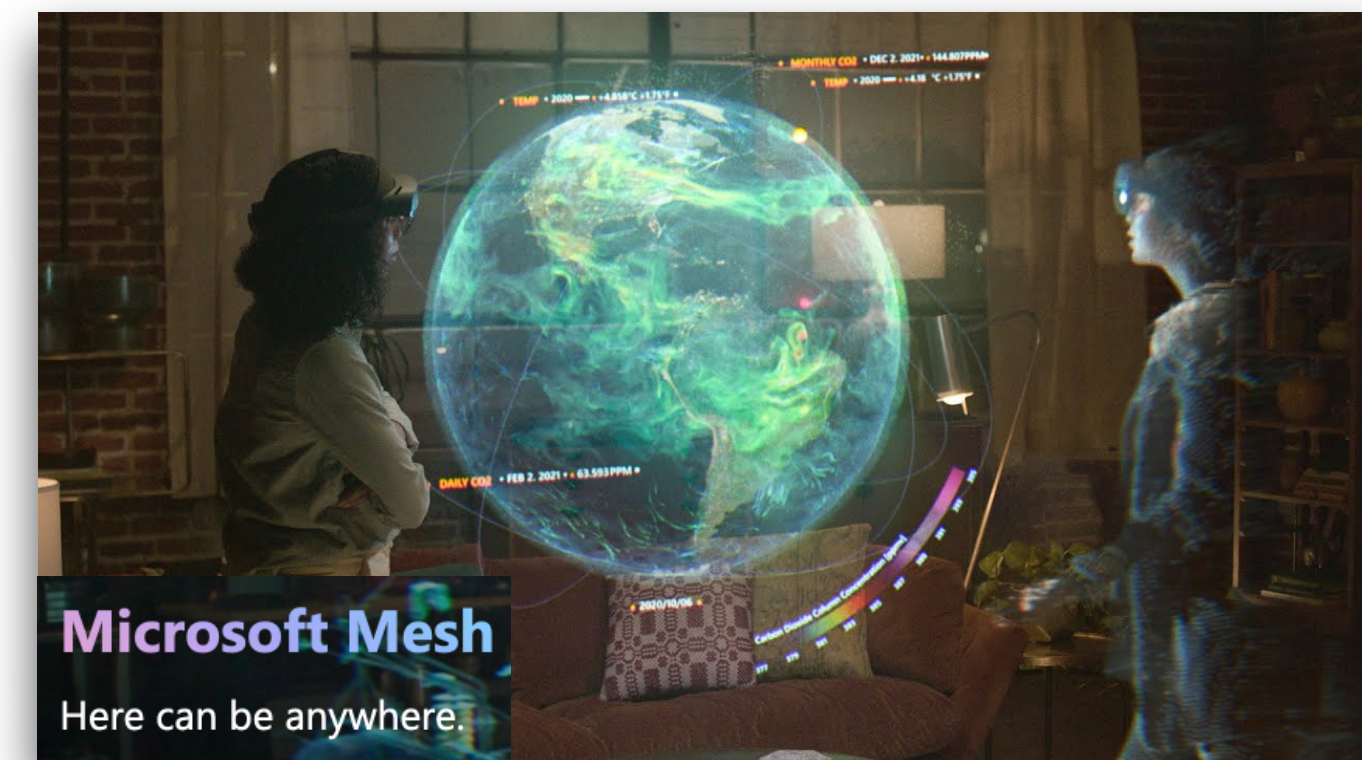
가상현실 소셜미디어 ‘호라이즌(Horizon)’



가상 오피스 ‘인피니트 오피스(Infinite Office)’

마이크로소프트(Microsoft)

디지털 트윈(Digital Twin)



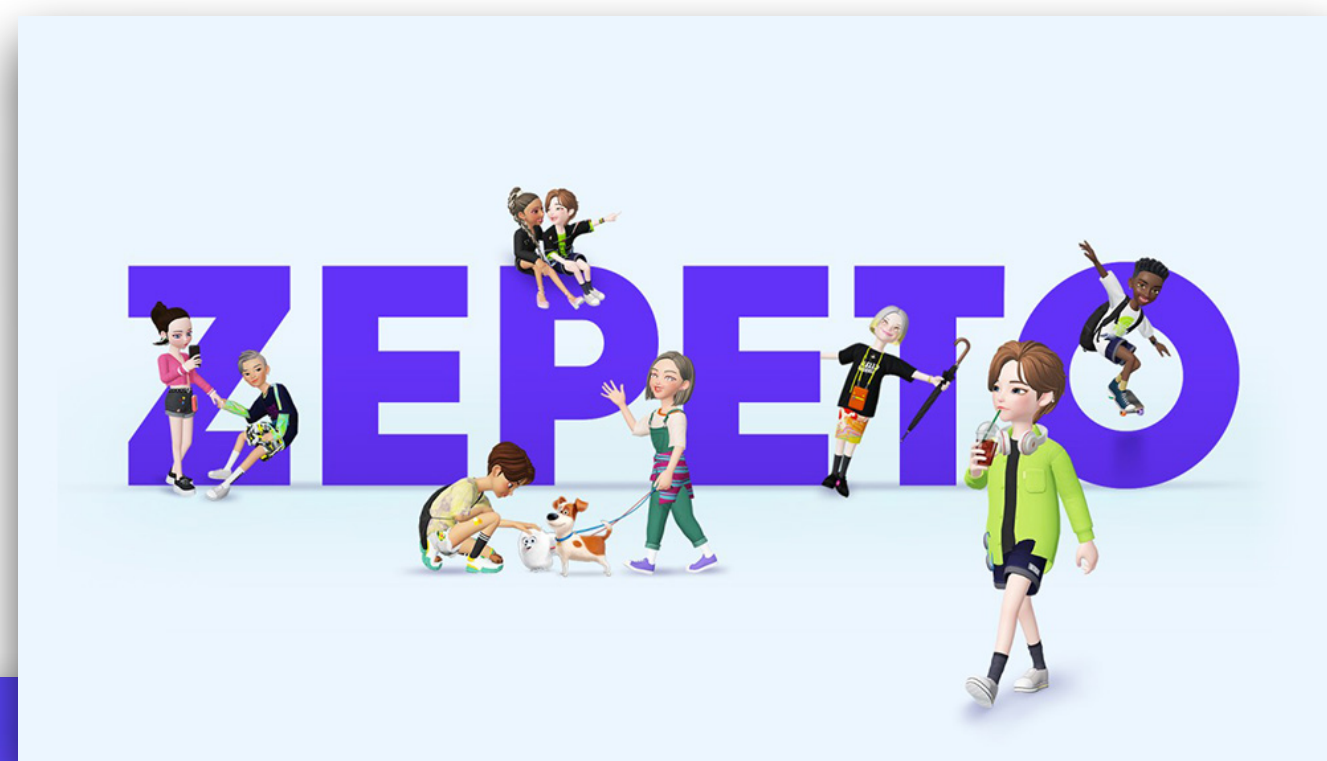
클라우드 기반 디지털 트윈 ‘메시(Mesh)’



가상과 현실을 ‘디지털 트윈’으로 동기화

네이버Z 제페토(ZEPETO)

아바타(콘텐츠) 기반 메타버스 플랫폼



ZEPETO
CREATE ITEM + SALES

제페토 아이템 디자인 따라 만들기!
수익 공개!



제페토 스튜디오에서
나만의 아이템을
만들어보세요

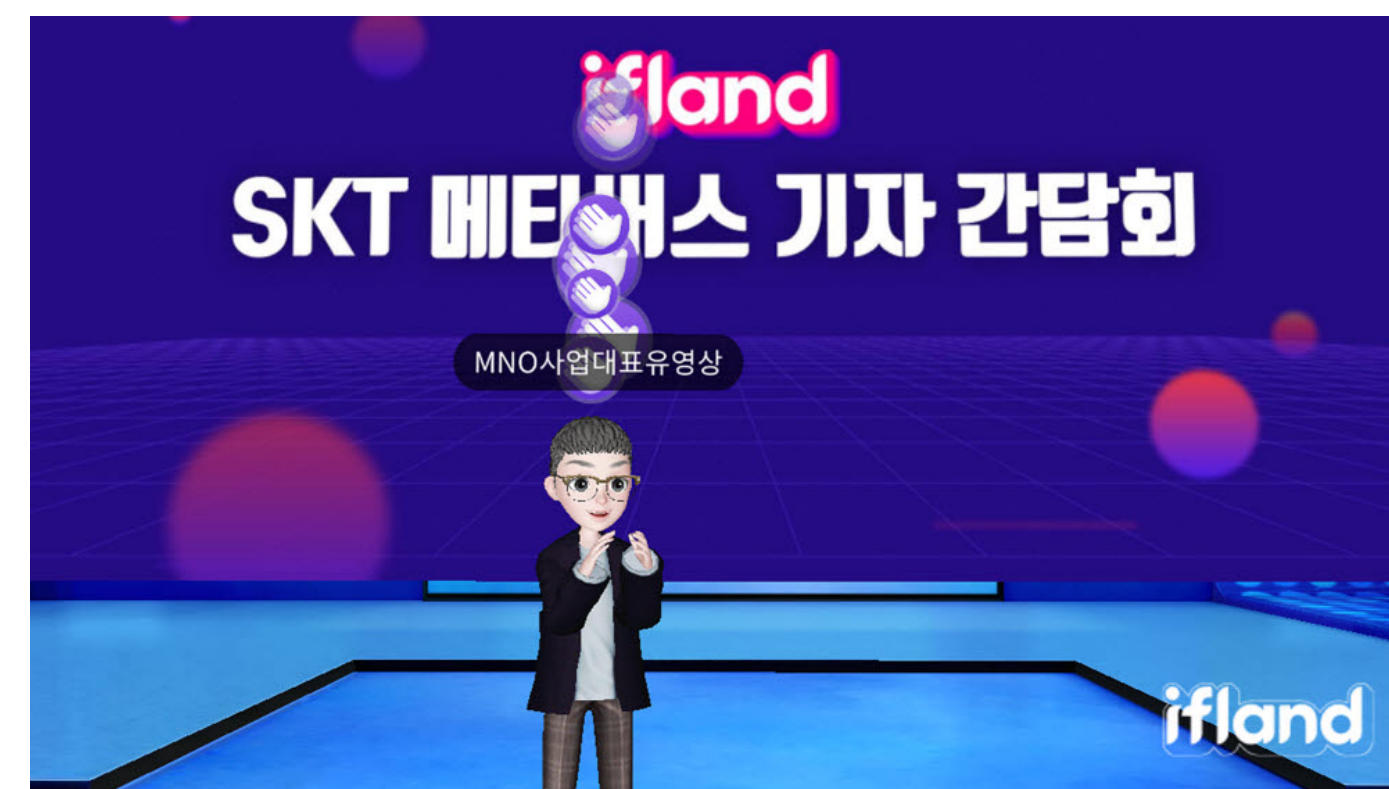
템플릿을 이용하여 쉽게 시작할 수 있어요

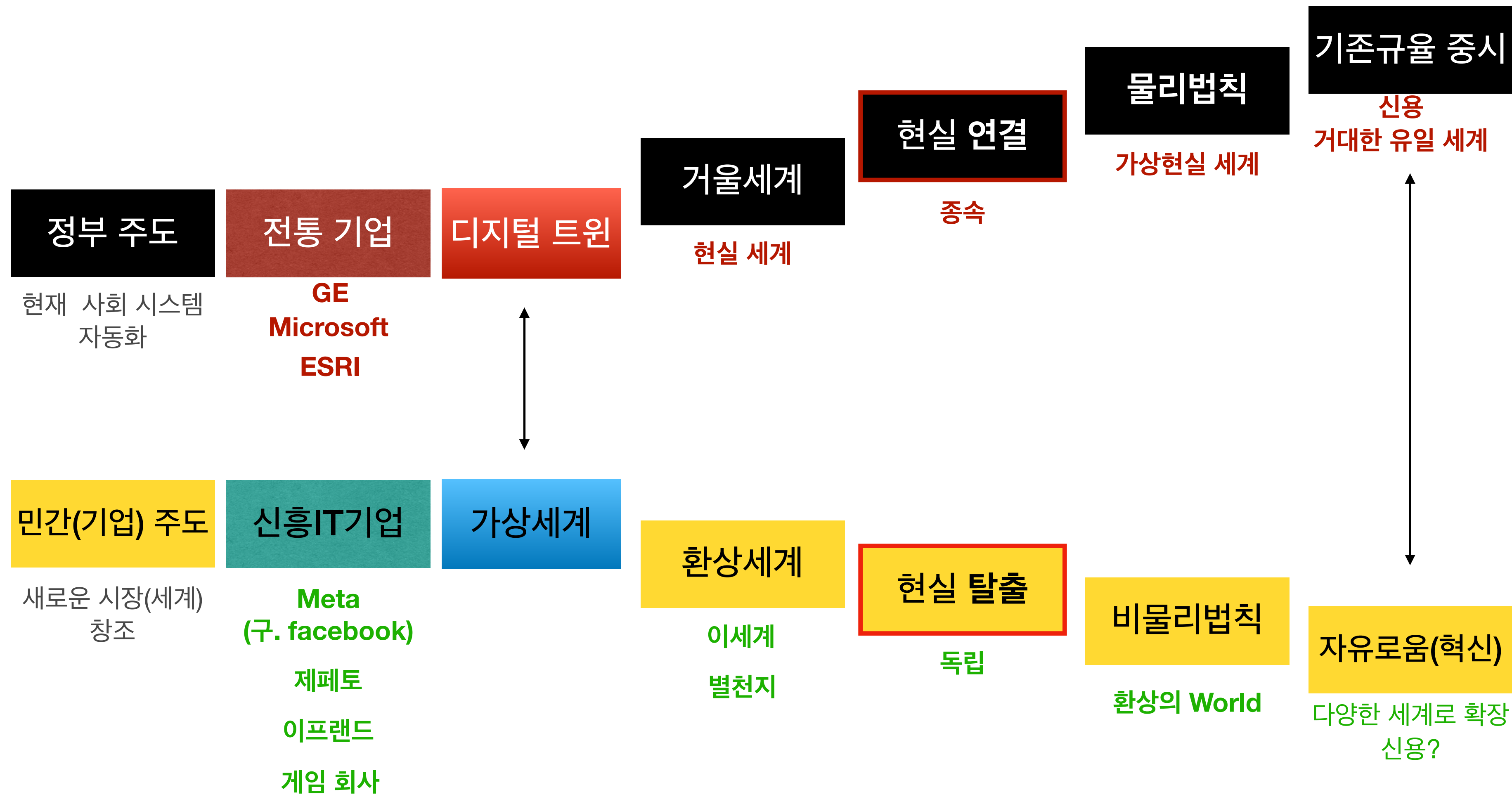
시작하기



SKT 이프랜드(ifland)

소통 모임에 특화된 메타버스 플랫폼





The Matrix



레디 플레이어 원
Sword Art Online

3. 해양정보 플랫폼



Digital Earth

Information Infrastructures

Citizens and Sensors

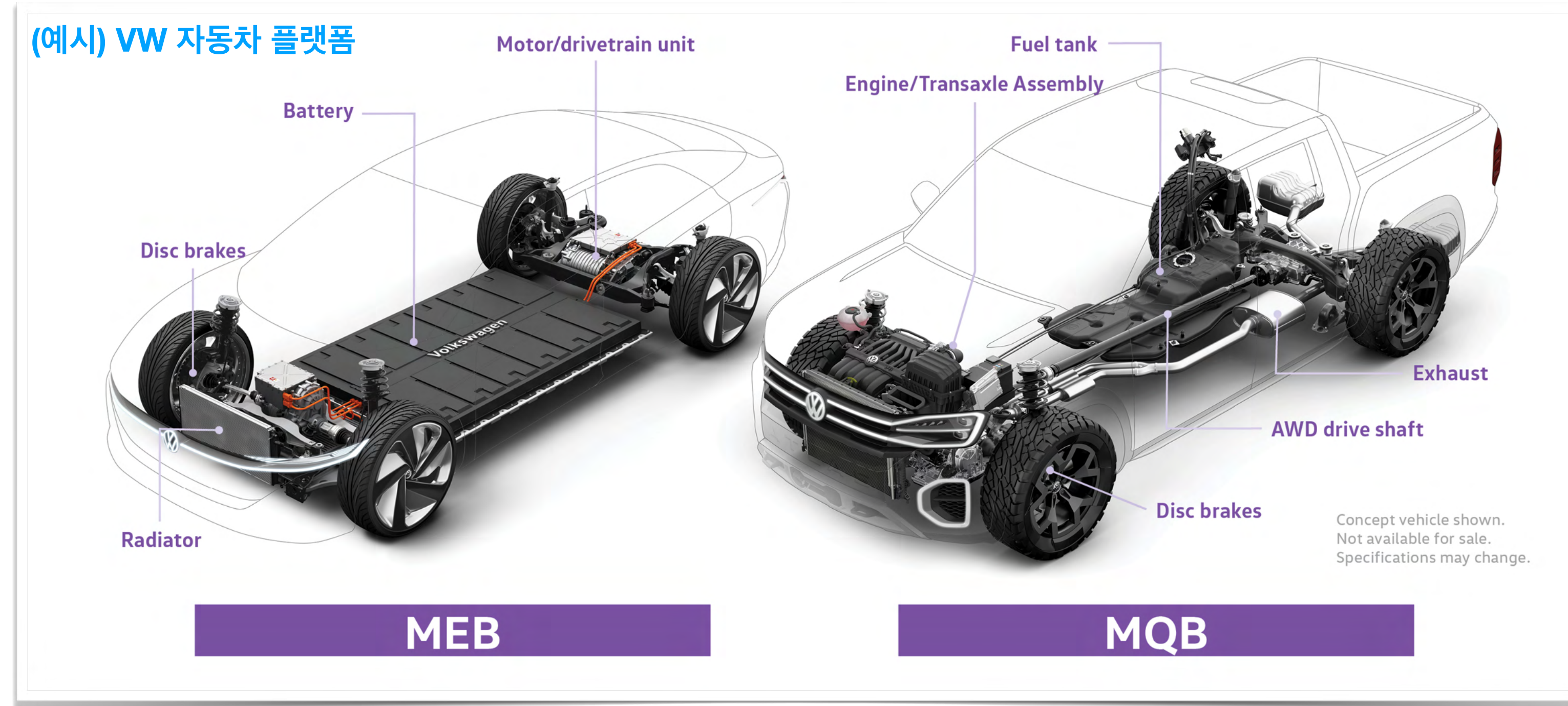
Bringing it all together

국내 해양데이터 플랫폼?

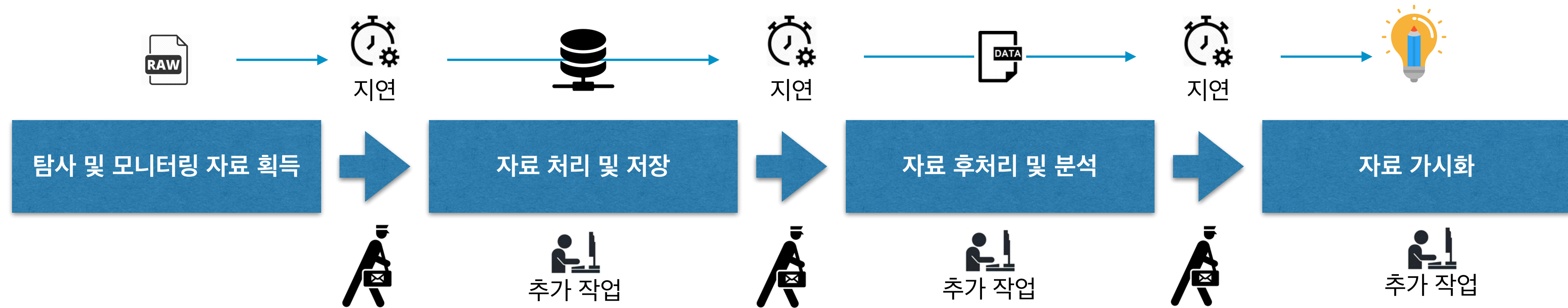
Ocean Data Platform?

플랫폼?

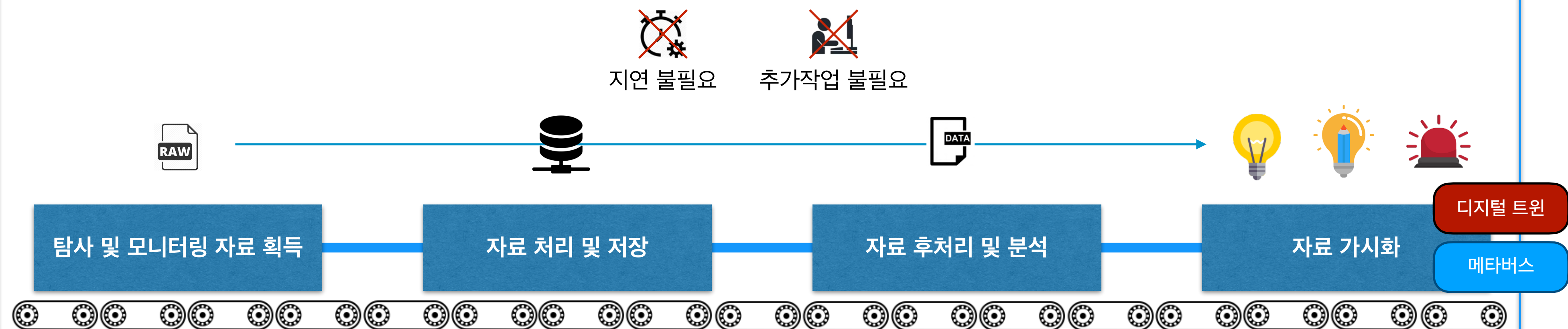
응용 기술 개발에 근간이 되는 핵심 기술들의 집합체 (group of technologies)



현재: 과정별 처리 (시간 & 작업 추가 소요)



미래: 실시간 처리 (자료 생산에서 분석까지 전과정 자동연결화)





European Centre for Medium-Range Weather Forecasts



Destination Earth (DestinE)

Digital Earth

Information Infrastructures

Citizens and Sensors

Bringing it all together

Destination? Earth's Digital Twin!



Source: European Commission

The European Union is currently building a digital twin of Earth that will irreversibly change the way we conduct climate forecasts.

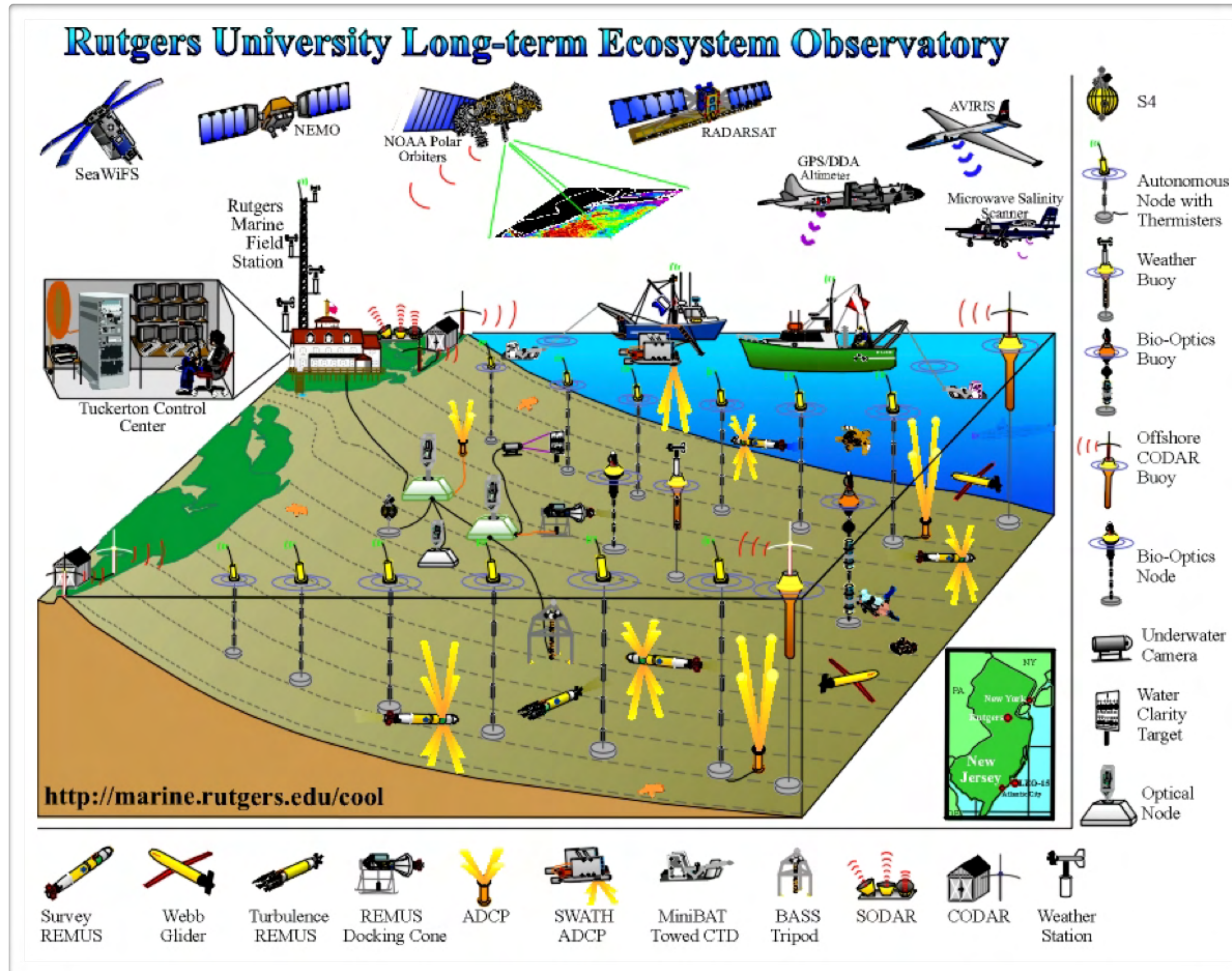
Earth's digital twin will simulate the ocean, ice, land and atmosphere with unparalleled precision enabling scientists to forecast droughts, floods and years from days to years in advance. The Destination Earth (DestinE) initiative will go further by attempting to capture human behaviour, which would allow world leaders to clearly see the impact of climate change, but also of national/European environmental policies in place.

DestinE is one of many initiatives promoted to contribute to European Commission overarching Green Deal and Digital Strategy. Further, with public datasets across Europe planned to be opened up, DestinE will also represent an important component of the European Strategy for Data.

The initiative will kick-off in 2021 and will be implemented gradually over the course of the next 7 to 10 years, by initially running on three supercomputers to be deployed in Italy, Spain and Finland. The initiative will be developed through the following milestones:

- In 2023: Launch of an operational cloud-based enabling platform and the first two digital twins.
- By 2025: Platform integrates 4-5 operational digital twins and offers services to public sector users for developing, monitoring and assessing the impact of the proposed policy and legislative measures concerning the environment and climate.
- By 2025-30: Development towards a full digital twin of the Earth through a convergence of the digital twins already offered through the platform.

Long-term Ecosystem Observatory



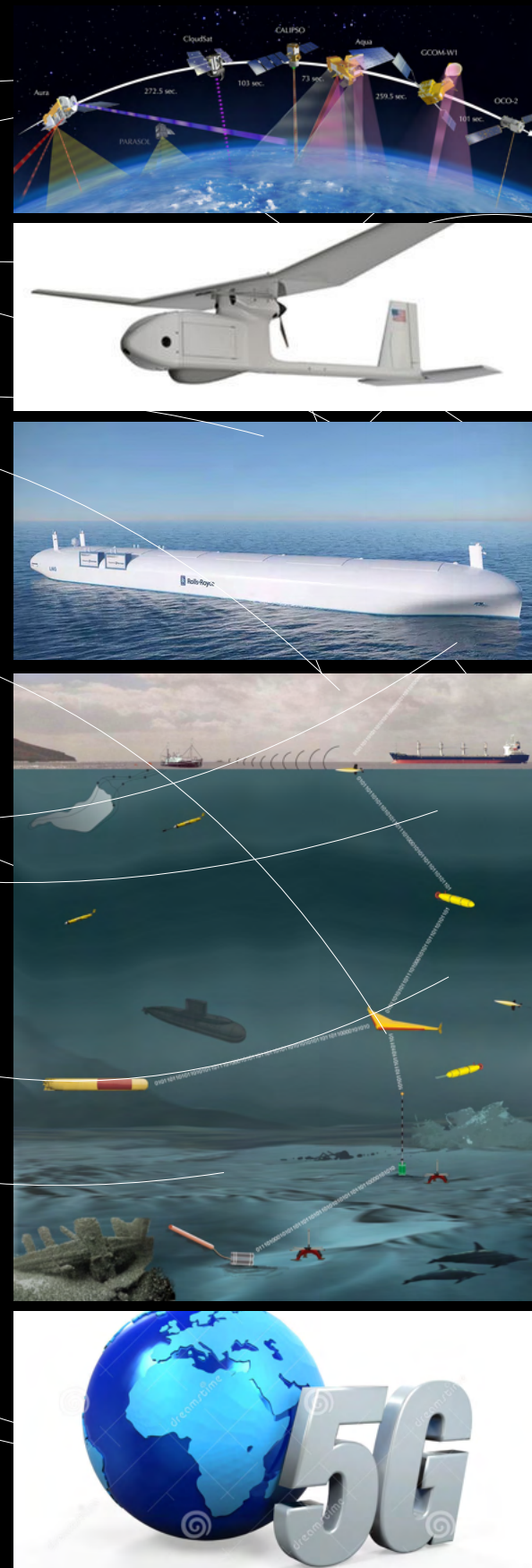
CYBER BLUE (KIMST, 2017)

(Digital Ocean: Smart Ocean-Science Research Platform)

Real World



Smart Ocean
Observation



Ocean Science Cloud
Computing Platform



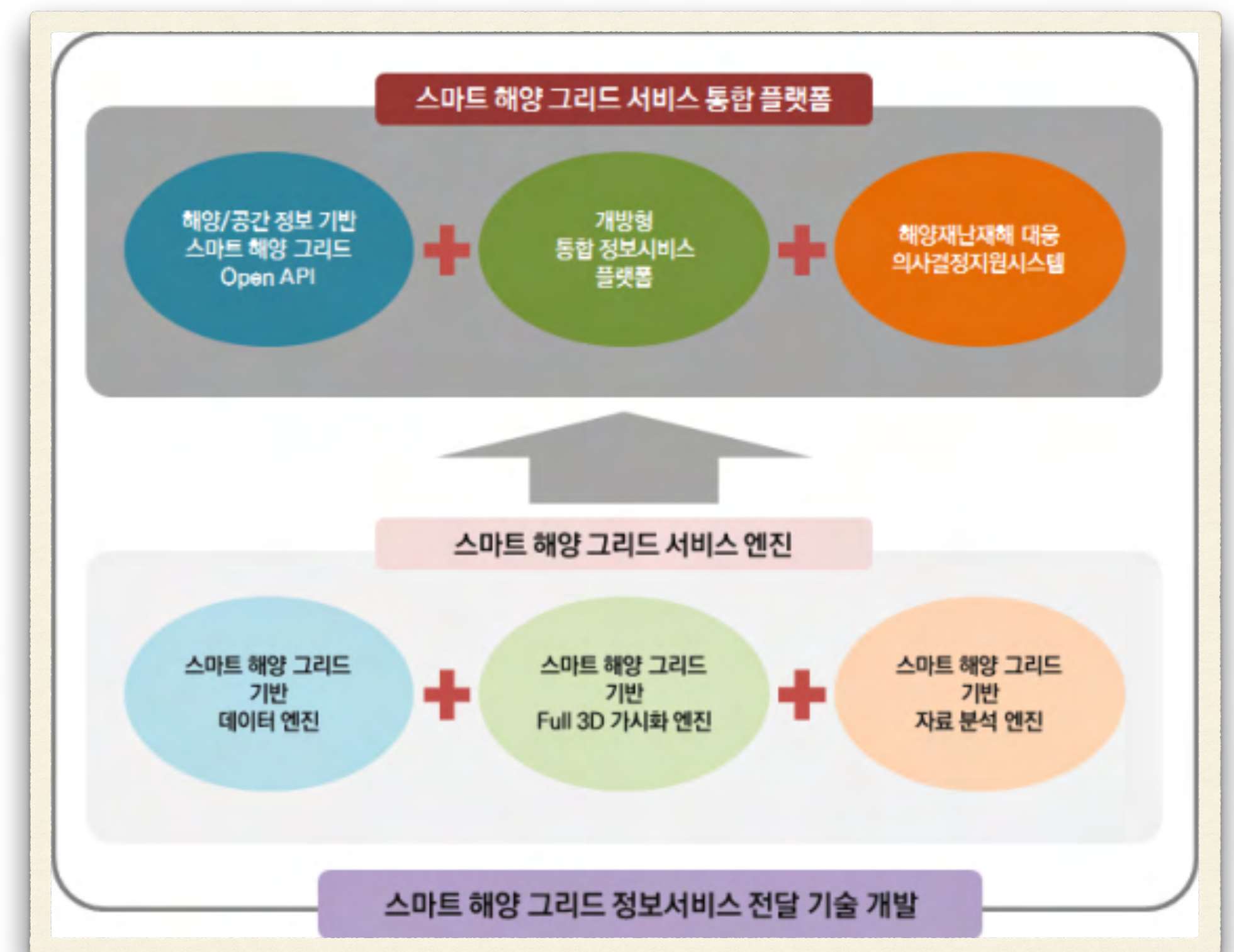
Virtual World



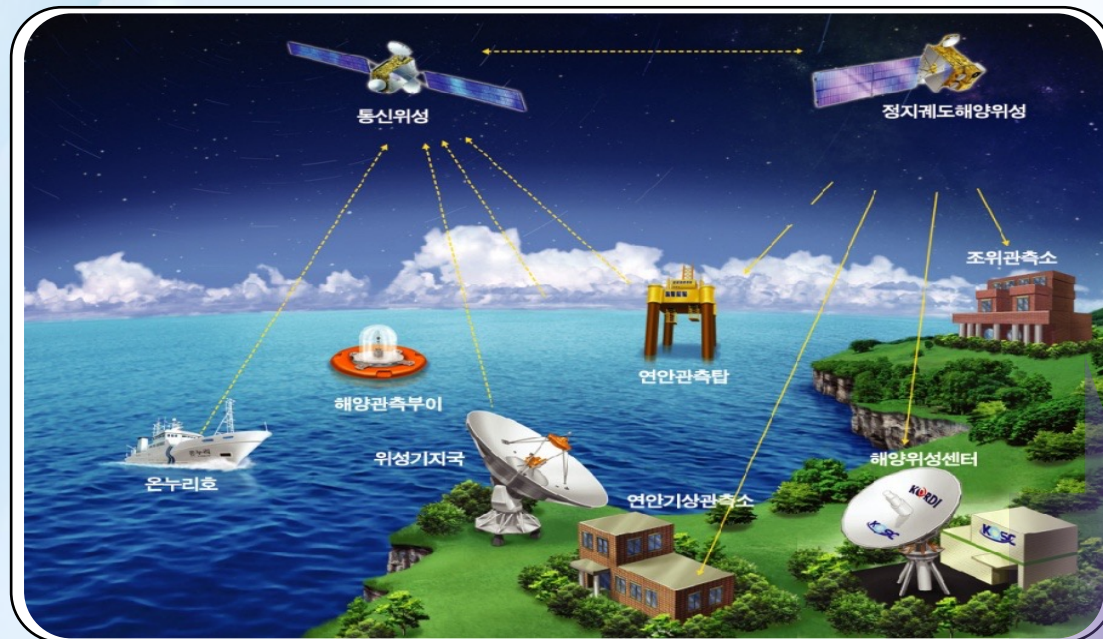
(예타기획 1차) 빅데이터 기반의 해양 변동 예측기술개발 ('18)

(예타기획 2차) 스마트 해양 그리드 정보서비스 전달 기술 개발 ('20)

무인 해양관측 정보 생산체계 구축 및 슈퍼컴·AI기술을 활용한 해양예측 정보 생산을 통해
해양정보 서비스 산업 창출



Big Data



Supercomputer (AI)



Information Service



Artificial Organ



3D Printer

(목표) 해양 관측 무인체 · 운영 기술확보 및 자료통합 체계 구축

- (세부과제1) 해양무인체(글라이더, 드론 등) 관측 및 통합 통신 기술 개발
- (세부과제2) ICT 융합 해양유 · 무인 장비 운영 플랫폼 및 자료통합 체계 개발

세부내용

○ (세부과제1) 해양무인체 관측 및 통합 통신 기술 개발

- 무인체 인공지능 자율운항 시스템 개발(자율운항 알고리즘 및 소프트웨어 등)
- 해양 관측자료 생산 센서 및 통신장비 등 해양환경 모니터링용 장비 개발
- 무인체 관측시스템간 실시간 고속 무선통신망 네트워크 구축

○ (세부과제2) ICT 융합 해양유 · 무인 장비 운영 플랫폼 및 자료통합 체계 개발

- 대용량 관측 데이터 실시간 전송 및 자동 자료처리를 위한 전송 기술 및 Big Data DB 구축
- 해양 관측자료 실시간 분석을 위한 딥러닝 알고리즘 개발

(목표) 해양환경 관측 Big Data분석 및 의사결정 모델링을 위한 초고성능컴퓨터 도입 및 운영체제 구축

- (세부과제1) 해양과학데이터입수·개방·활용·관리체계구축
- (세부과제2) 해양과학 초고성능컴퓨팅 전문센터 구축 및 활용

세부내용

○ (세부과제1) 해양과학데이터입수·개방·활용·관리체계구축

- 대용량데이터 실시간 수집체계 구축 및 데이터 분석·처리 SW 개발
- DOI 기반 데이터 재활용 및 보호체계 구축

○ (세부과제2) 해양과학 초고성능컴퓨팅 전문센터 구축 및 활용

- Pfllops 규모 초고성능컴퓨팅 시스템 구축
- 초고성능컴퓨팅 운영을 위한 기반시설 (UPS, 축전지, 발전기, 자동제어장비, 방재장비 등) 구축
- 유전체, 해양예보 등 서비스별 데이터 가시화 SW 개발 및 통합포털 구축/운영
- 초고성능컴퓨팅 자료분석 및 활용 전문인력 양성

3 추진과제 3 (AI기술을 활용한 해양환경 예측시스템 구축)

디지털 트윈

메타버스

(목표) AI기술 활용 해양환경 예측시스템 설계 및 정보서비스 창출

- (세부과제) 인공지능 4D 디지털 해양환경 공간정보 구현 및 정보서비스 공급체계 구축

세부내용

○ (세부과제) 인공지능 4D 디지털 해양환경 공간정보 구현 및 정보서비스 공급체계 구축

- 무인관측 Big Data 및 초고성능컴퓨팅을 활용한 실시간 분석 및 의사결정 시스템 개발
- 수요자 맞춤형 데이터 제공 플랫폼 서비스(Web, Mobile 등) 개발 및 산업지원 제도마련



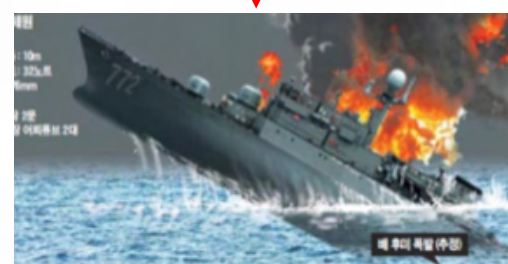
Applied Forecast



Oil Spills



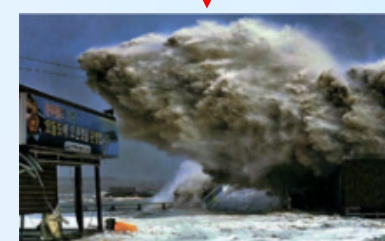
Red tides



Search and Rescue



Flooding/Inundation



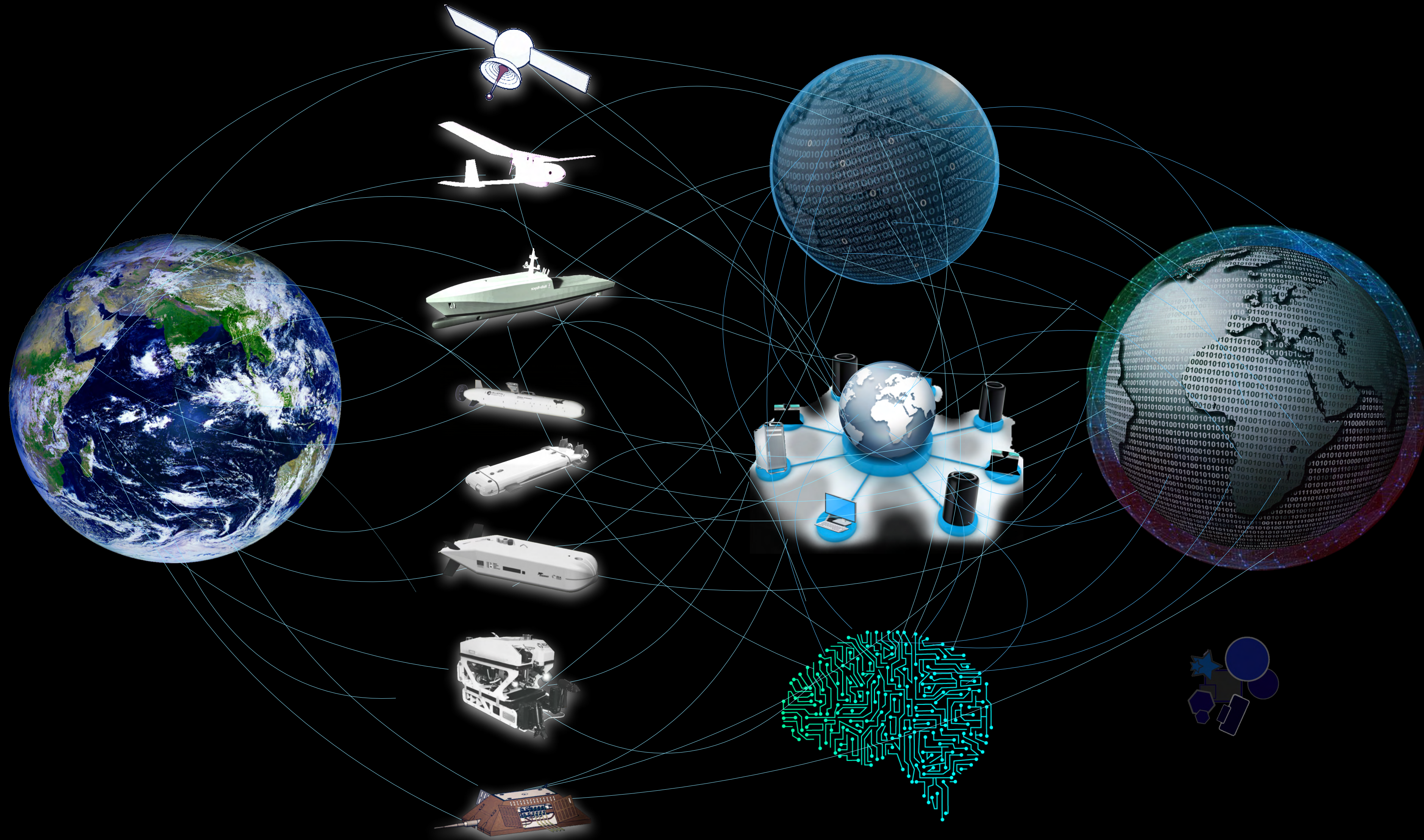
Storm waves



Korean PORTS

Leisure
Tourism
etc.

해양데이터 플랫폼



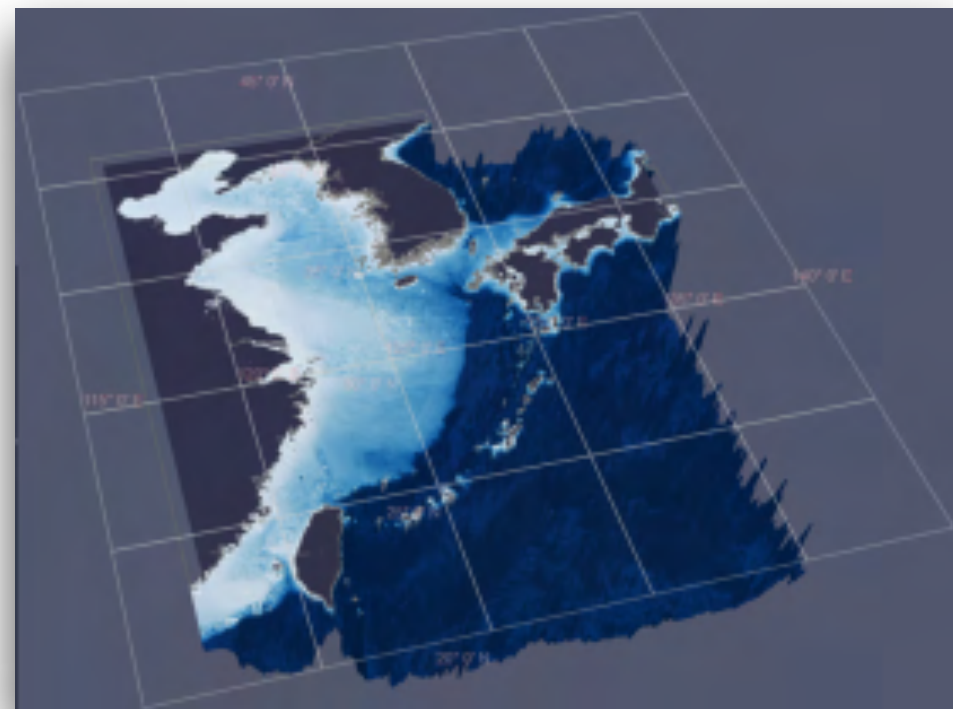
4. 해양 메타버스 현황

국내 해양분야 메타버스 관련 기획, 연구, 기업? ...

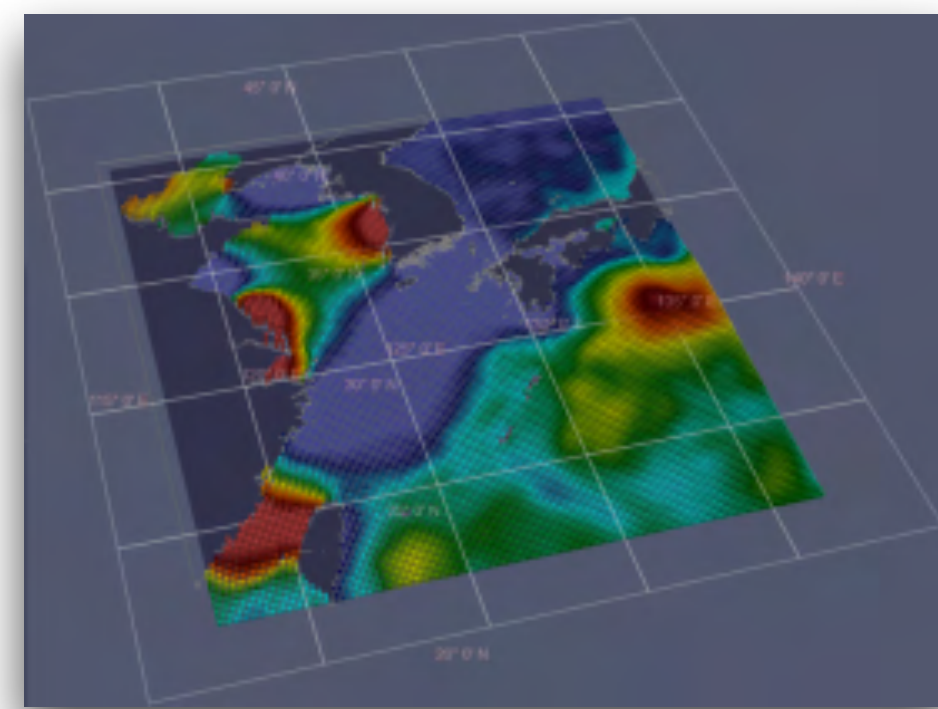
블루 오션

(주)비엔티 - 해양정보 관련 3차원 가시화 엔진(Ocean3D-FX) 개발 중

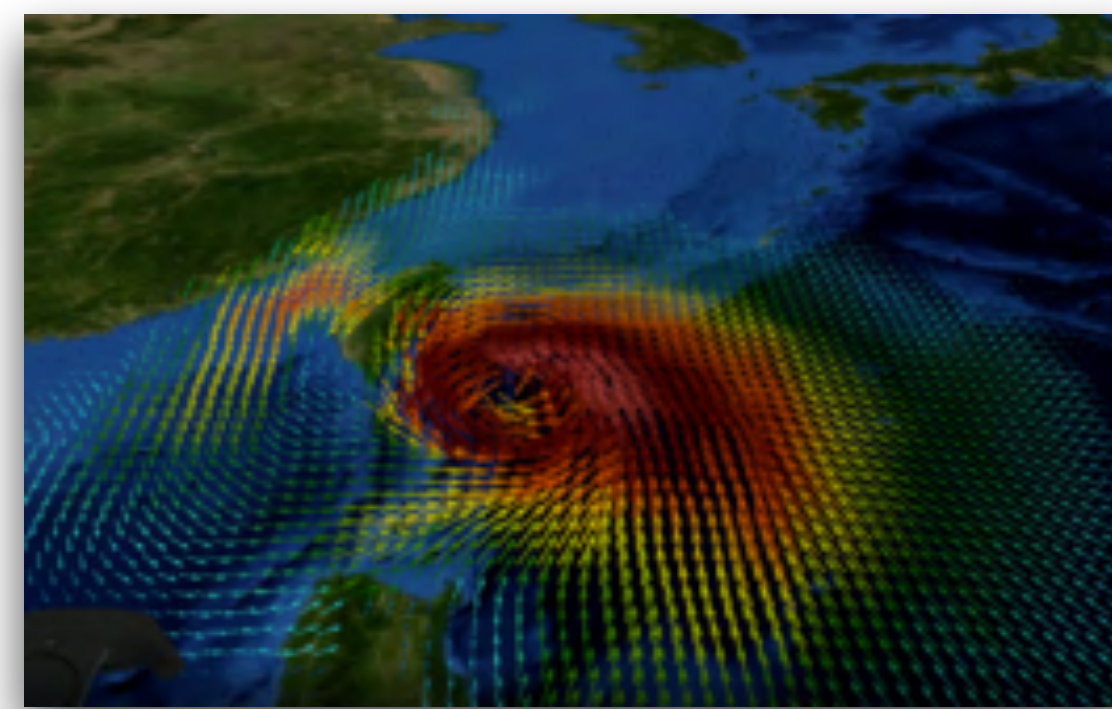
Ocean3D-FX



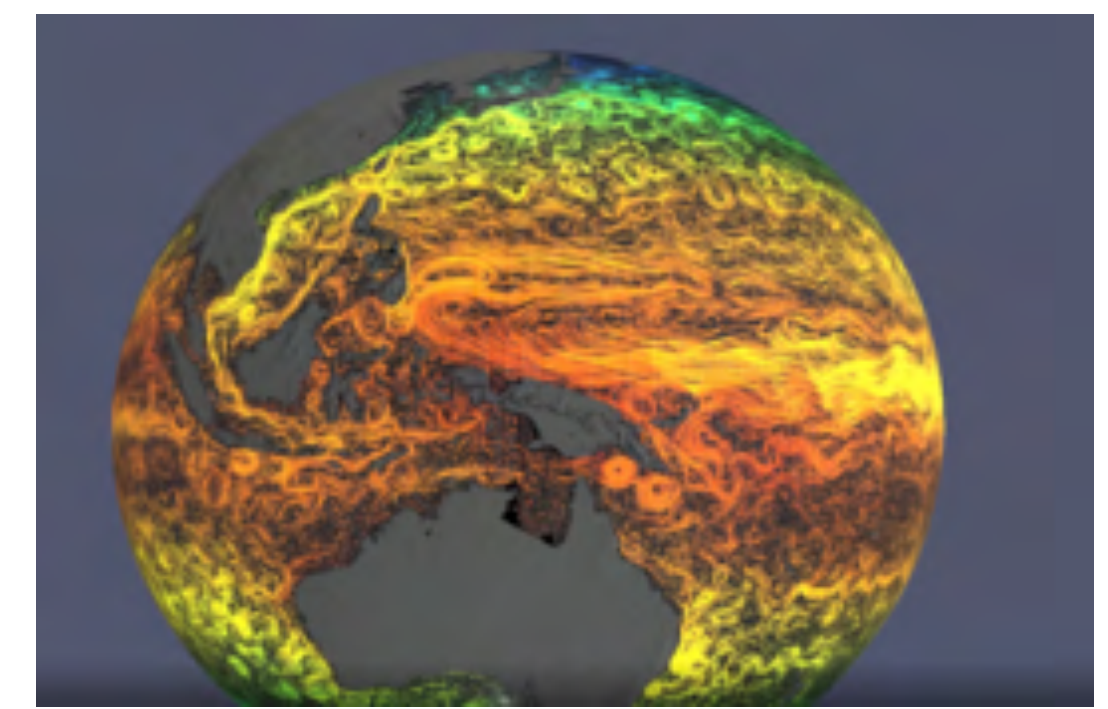
해저지형



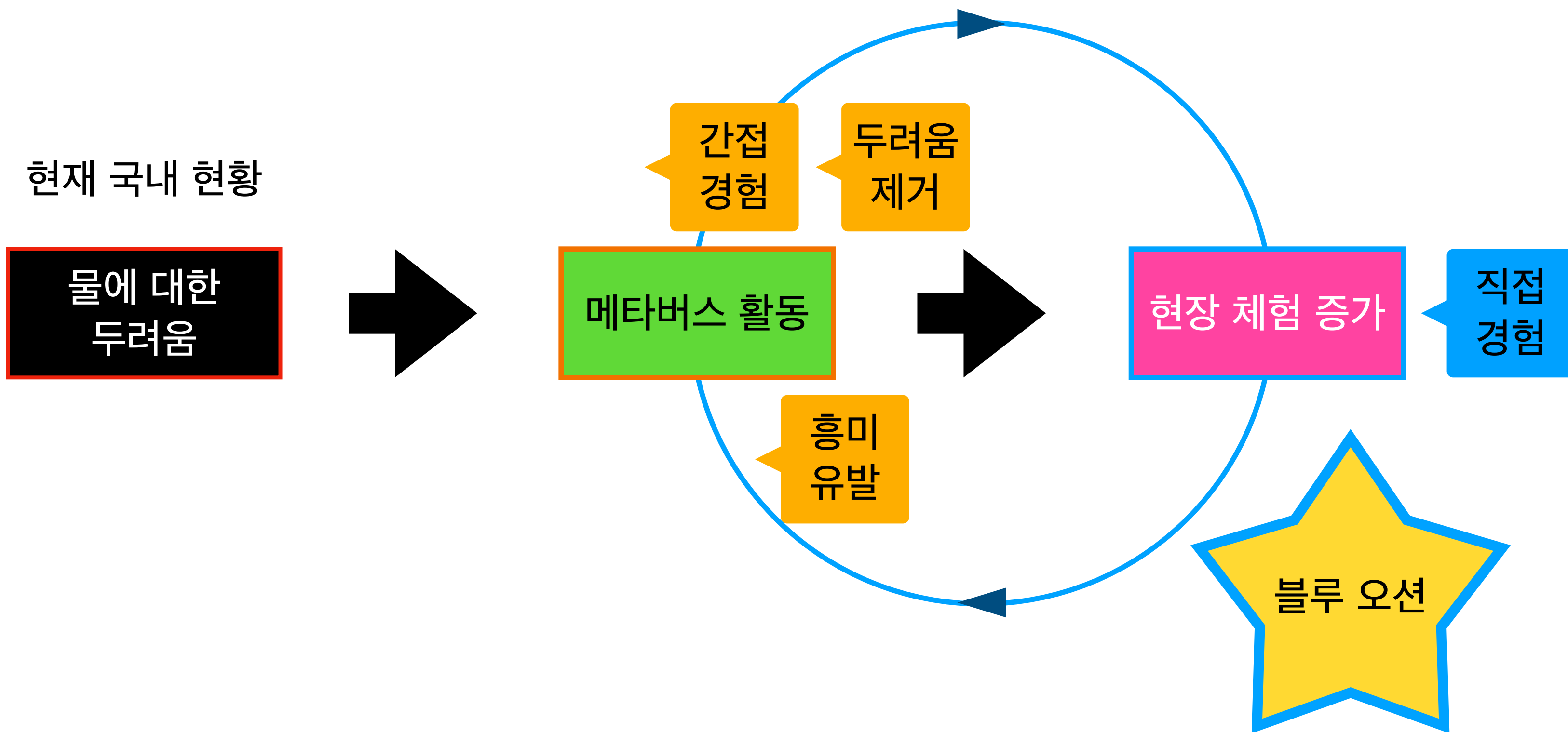
해수면



한반도 주변 해역 해류



전 지구 해류



해양 데이터 사냥꾼?

DatacenterDynamics

The Business of Data Centers.

Data is money: stand by for the Gold Rush

25 November 2016 By Dale Green, Digital Realty

The gold rush is on again, and once again it is centered on the American West. The original gold rush started at [Sutter's Mill](https://en.wikipedia.org/wiki/Sutter's_Mill) (URL=https://en.wikipedia.org/wiki/Sutter's_Mill) in Coloma. This time round, it is starting in Silicon Valley, and you will need more than a pan and a slice of luck to find your fortune.

With the big data revolution showing no signs of slowing down, many companies – and more than a few governments – are recognising that it is data itself that will drive the tech products of tomorrow, and are hoovering up vast amounts of the stuff. The monetizing of this data is, for some, becoming a 21st-century gold rush.

Data is Currency. Invest Wisely

Unlocking the value of big data requires a proactive approach to deriving insights. That's where data analytics tools can help

For example, Siemens PLM Software is helping customers save millions of dollars and improve product performance. Their innovative technology (which runs on Dell gear) takes a big data approach that searches billions of data records in seconds and helps pinpoint emerging issues within the supply chain. Read the case study.

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The value of data

Accenture Interactive – Point of View Series

What's your data worth?

Gain insight into revenue opportunities
embedded in enterprise data

DATA IS THE NEW CURRENCY

The simplest cell phone call is a miniature data gold mine. Two people talking on their mobiles think they are having a simple conversation, but as they chat, data is being collected about their location, the time and length of their call, and possibly the way that information interfaces with their other digital activities.

MasterCard, she works to ensure that the company maintains an open and respectful relationship with the data of its customers.

"My job is to be the voice of the individual," she says. "We always start with the individual. What are our customers or cardholders going to experience? What are they going to see? Do

Stonier does not like to be managed herself, so at MasterCard, she works to ensure that the company maintains an open and respectful relationship with the data of its customers.



JOANN STONIER

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MIT Sloan
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What's Your Data Worth?

Magazine: Spring 2017 Issue • Research Highlight • March 23, 2017 • Reading Time: 9 min

James E. Short and Steve Todd

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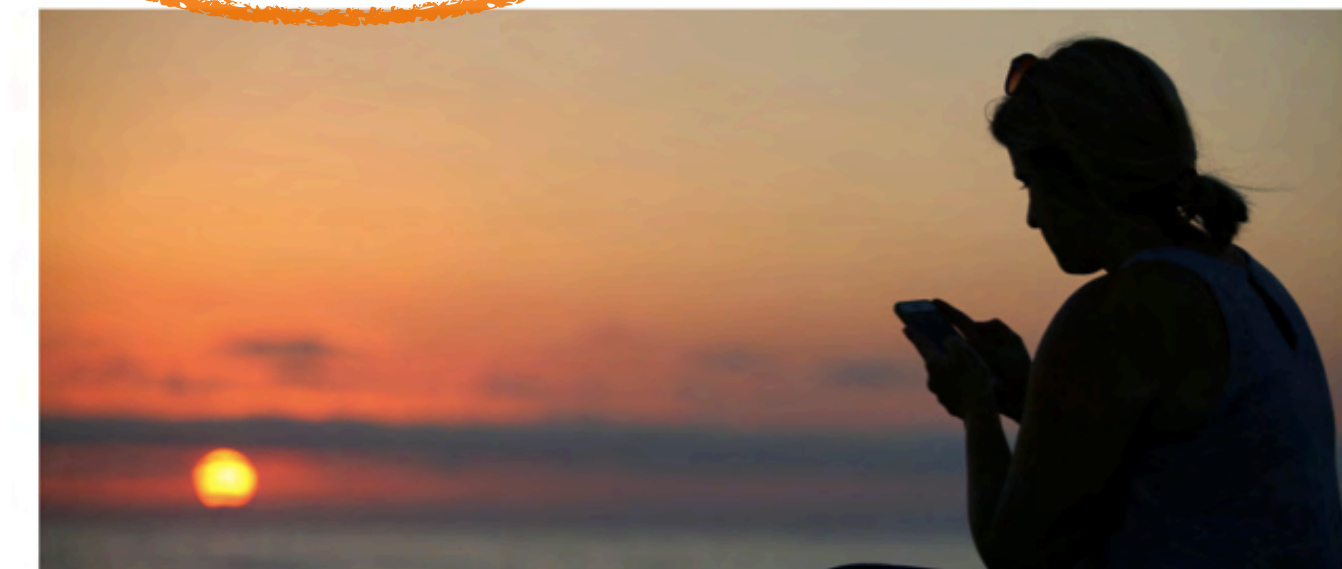
POST WRITTEN BY

Craig Strong

Chief Technology Officer for Insight Software Ltd (GoHubble.com).

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The value of data



Who will be the winners and losers as the age of open data takes off?
Image: REUTERS/Mike Blake

22 Sep 2017

Vasudha Thirani

Policy Associate, #DigitalIndia Foundation

Arvind Gupta

Chief Technology Officer, Bharatiya Janata Party (BJP)

국가전략, 제26권 2호 2020년 여름호, 5~34.

데이터 안보와 디지털 패권경쟁: 신흥안보와 복합지정학의 시각*)

김 상 배**

MK 뉴스

인쇄하기

취소

중국, 미국의 IT 압박에 '데이터 안보' 자체 구상으로 반격

'글로벌 데이터 안보' 이니셔티브로 미국 견제 우군 확보전
"중국, 데이터 보호 철저...해외 중국기업에 위법 요구안해"

| 입력 : 2020.09.08 11:37:37

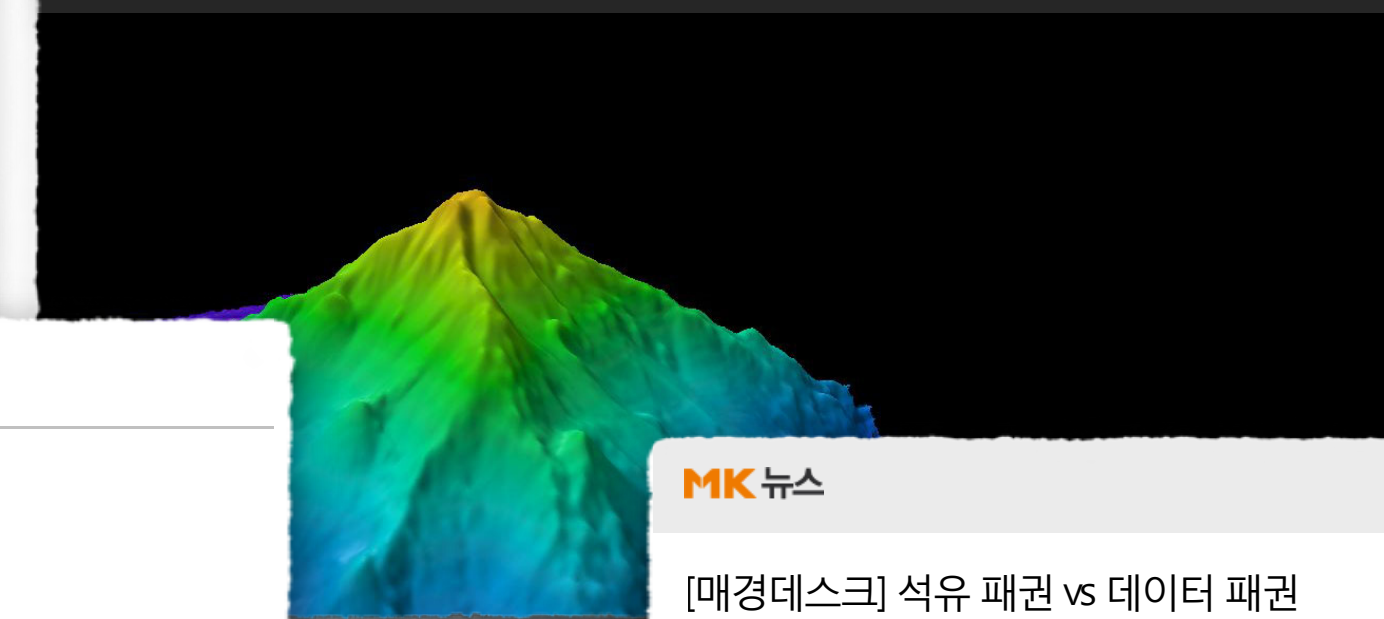
요약

최근 세계 주요국들이 벌이는 디지털 패권 경쟁의 과정에서 '데이터'의 중요성이 강조되고 있다. 특히 데이터 문제를 국가안보의 관점에서 이해하기에 이르렀다. 그러나 오늘날 데이터의 동맹과 연대외교의 쟁점이 되었으며, 정보기관 네트워크나 군사 정찰위성을 통한 데이터 수집 활동의 중요성도 커졌다. 전통 군사안보 분야에서도 비데이터 역량은 미래적 수해

Trump plan to push seafloor mapping wins warm reception

White House orders agencies to come up with plan to accelerate efforts

22 NOV 2019 · BY WARREN CORNWALL



MK 뉴스

인쇄하기

취소

[매경데스크] 석유 패권 vs 데이터 패권

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민관 투자와 생활권 경쟁
제조강국 '기적' 일궈내

'유한' 석유보다 파괴적인
'무한' 데이터 시대 진입
누가 큰 그림을 그리고 있나

황형규 기자 | 입력 : 2021.10.11 00:07:01

연합뉴스

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| 홍콩매체 "중국 빅테크부터 스타트업까지, 메타버스 사업 뛰어들어"



페이스북의 새로운 사명 '메타'와 새 로고
(엔로파크 로이더=연합뉴스) 지난 10월28일(현지시간) 미국 캘리포니아 엔로파크의 메타(옛 페이스북) 본사 앞에 새로 설치된
류한석의 진기술 토크
간판. 2021.11.2 photo@yna.co.kr

(홍콩=연합뉴스)
싼 국가안보 위

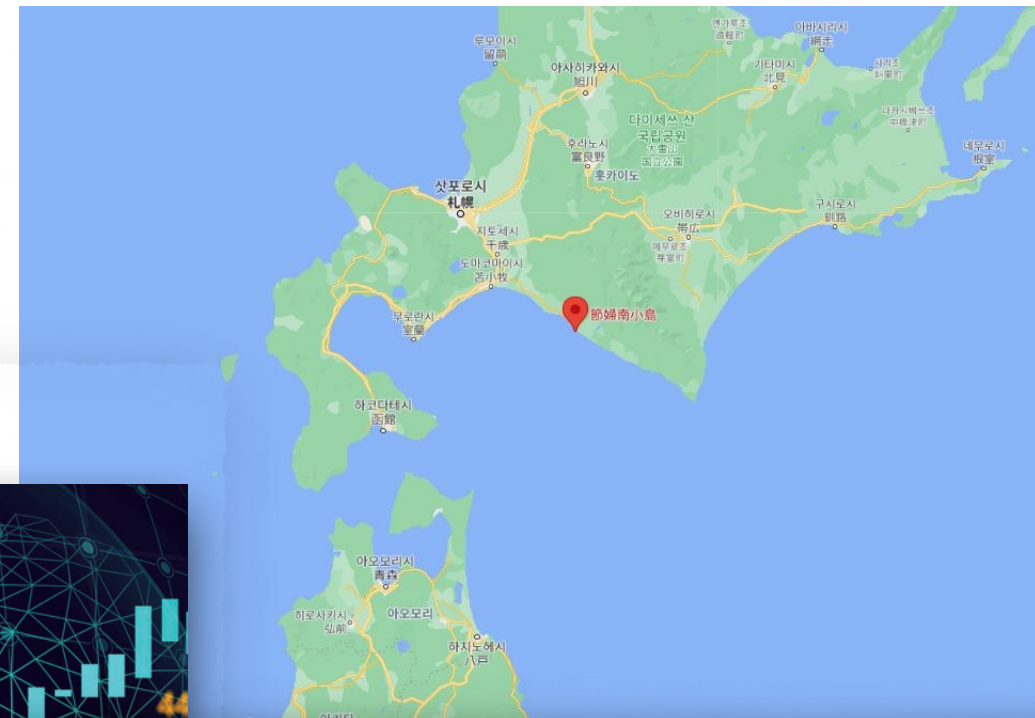
데이터경제를 향한 글로벌 패권 경쟁

연합뉴스

일본, 영해 확보 '안간힘'...'측량 항공기로 동해 섬 조사'

송고시간 | 2021-08-21 11:51

| 침식 등으로 소실 가능성 있는 섬 조사 착수...EEZ에도 영향

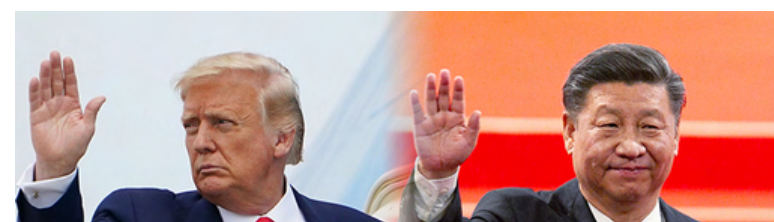


세계일보

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각국과 접촉... 美 견제 우군 확보

입력 2020-09-08 14:29:12, 수정 2020-09-08 19:27:23



10/23/21, 7:56 PM Mapping quest edges past 20% of global ocean floor - BBC News

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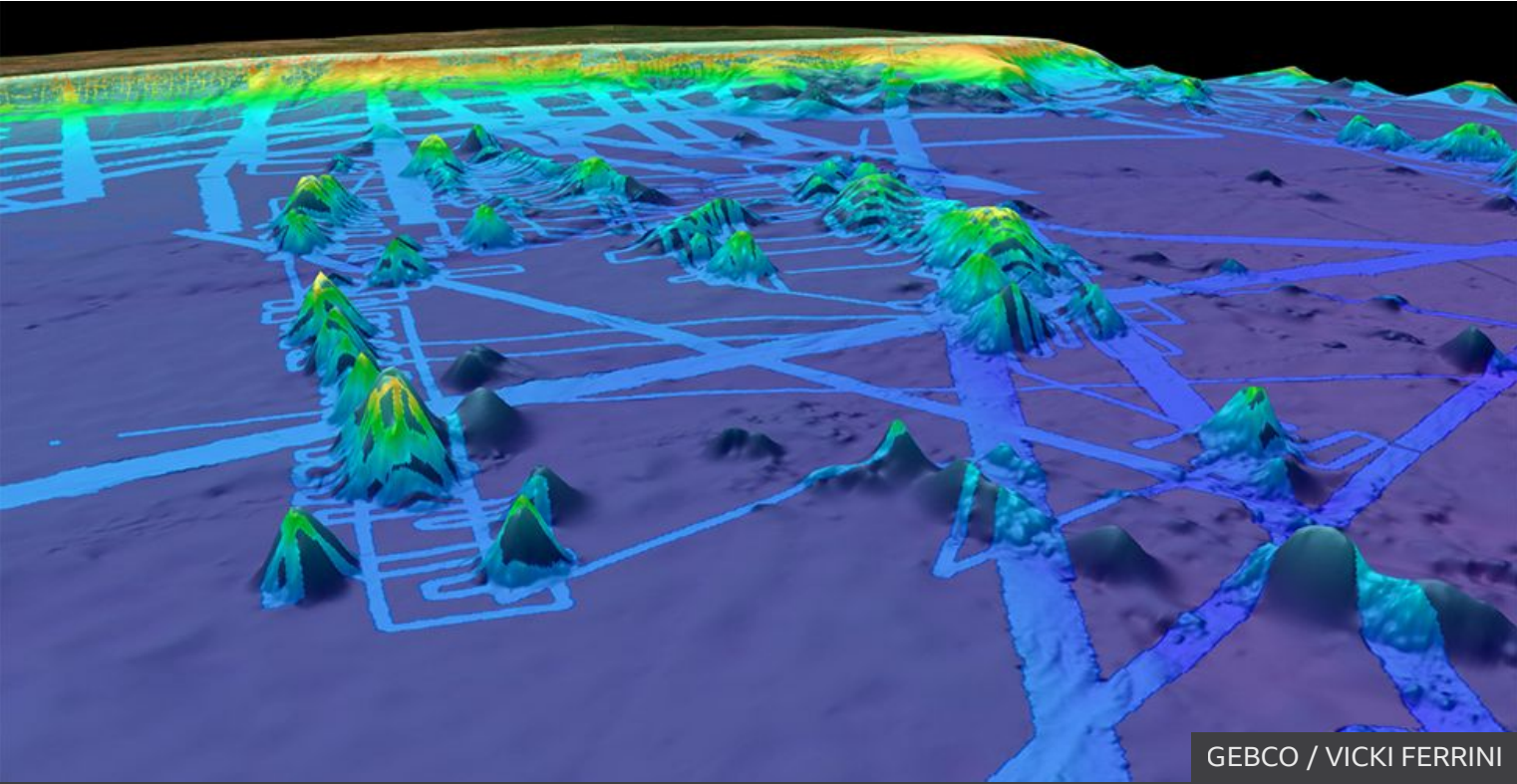
Mapping quest edges past 20% of global ocean floor



Jonathan Amos
Science correspondent
@BBCAmos

21 June





GEBCO / VICKI FERRINI

Seamounts off Brazil: It's around underwater mountains that marine life will congregate

The quest to compile the definitive map of Earth's ocean floor has edged a little nearer to completion.

Modern measurements of the depth and shape of the seabed now encompass 20.6% of the total area under water.

ars TECHNICA

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DRONE ON —

XPRIZE selects winners of autonomous seafloor-mapping competition

Over \$7 million goes to finalists who built the best systems.

SCOTT K. JOHNSON - 6/2/2019, 3:40 AM



Enlarge / Part of the winning team's system.

The winners of the latest XPRIZE competition were announced on Friday. Sponsored by Shell with a bonus prize from the US National Oceanic and Atmospheric Administration (NOAA), the competition focused on the challenge of exploring the seafloor and saw 32 teams from around the world competing for \$7 million.

해양 데이터 생산·제공 산업 활성화 기대

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해양 레저



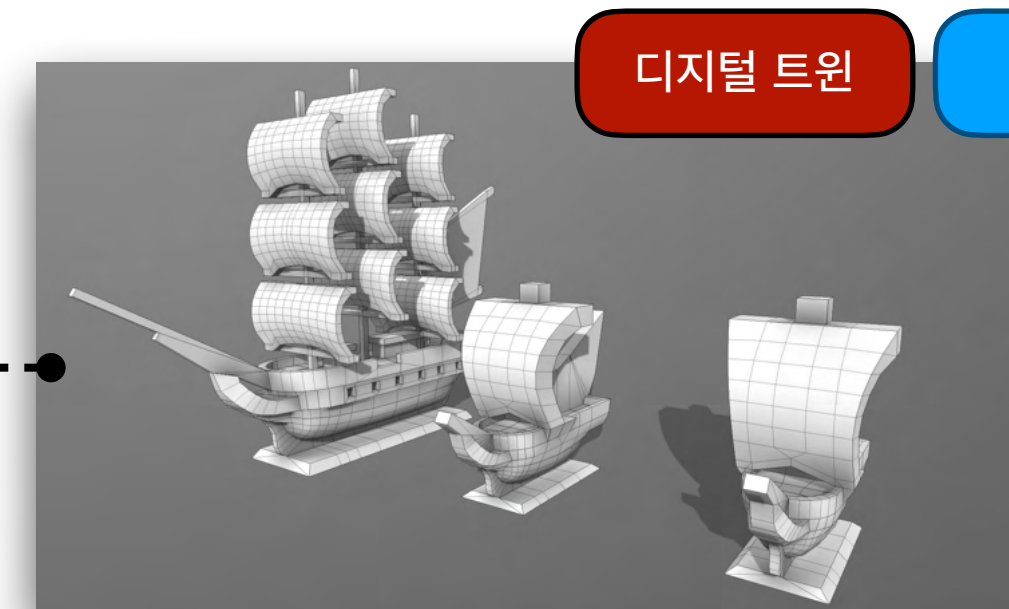
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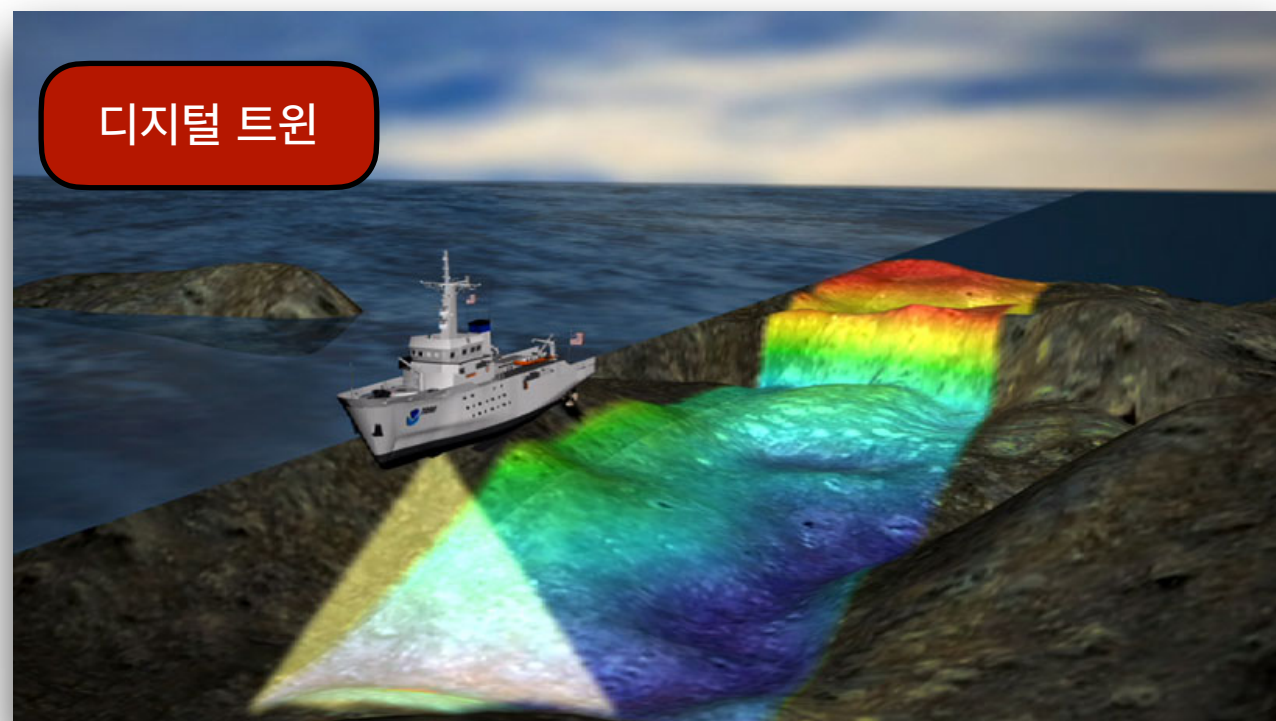
해양 디지털 트윈 & 가상세계



디지털 트윈

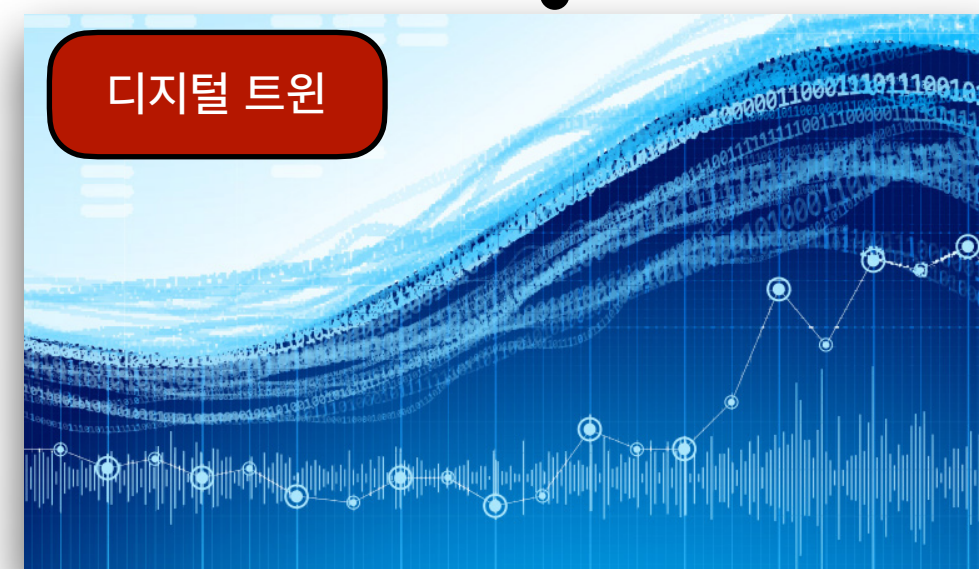
메타버스

가상 해양 선박/인프라 3D 구현/매매



디지털 트윈

해양 탐사 산업



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